SCI-RTC

Paired Raw-Timestream Conditioning and Temporal Response

Engineering Conformance View

Package version	v0.1
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Authority	Shared normative core in `src/common/`
View-specific content	Informative evidence and review guidance only
Status	Implementation-blind author draft for scientific-owner review

Claim boundary. This view does not assert scientific approval, implementation conformity, representation fidelity, validation, observational performance, or production readiness. It contains no independent normative science.

How to use this view

The six-file shared normative core below is imported exactly once and is the complete formal authority companion to the separate twelve-section science-team rationale. The rationale explains this authority but does not duplicate it. The material after the end-of-core marker is an informative method for recording evidence against that authority. It may not waive a requirement, select an owner decision, redefine an unavailable state, or restate an equation with different semantics.

A conformance review should create one evidence record per requirement and one falsification record per applicable prediction. Each record must bind an exact implementation and artifact identity, state what was inspected or executed, preserve unexpected error-level messages, and distinguish pass, fail, not exercised, blocked by owner decision, and unavailable evidence. Passing a fixture supports only the named requirement and claim layer.

> **Stop rule.** If a required owner decision is open, the affected operation or claim remains unavailable. Engineering must not choose a numerical value from current behavior and then describe it as scientific authority.

Contents

1 Shared normative authority and notation

Normative core stamp: `SCI-RTC-v0.1/r0.9`. The six files in `src/common/` form one authority. Text outside this shared core is explanatory or conformance guidance and cannot alter a definition, equation, assumption, requirement, prediction, unavailable state, or owner decision.

SCI-RTC begins with admitted aligned paired raw detector-coordinate streams `xs` and `rs` and ends with an atomic, consumer-neutral raw conditioning bundle. Conditioned x is the required numerical RTC output; the exact paired raw r parent and every causal r diagnostic or selector remain in that bundle. A conditioned r product is not required and needs a separately authorized channel-specific operator. Raw x is not itself a Stokes parameter. ALIGN, the Tune/readout mapping authority, AST, and SCI-BEAM retain authority for the upstream meanings they produce; SCI-CAL may calibrate only the conditioned x path of an exact RTC bundle. PTC, VAL, MAP, BEAM, NOI, MAP-003, FLT, SRC/MODE, and FRUIT retain authority for their downstream science. This contract owns the realized conditioning order and its scientific consequences; it does not infer current software behavior.

The words **shall** and **shall not** identify requirements. **Required unavailable** means that the affected product or claim shall not be reported as complete. **Optional unavailable** is a typed, scientifically inert state. Missing, disabled, automatic, rejected, invalid, failed, unavailable, unsupported, and omitted remain distinct states.

1.1 Indices, sets, and application-context labels

Symbol	Normative meaning
o	Immutable observation identity.
$d, q \in \mathcal{D}_o$	Exact admitted detector occurrences; d is a target and q may be a donor. A detector index or row number alone is not this identity.
m_{native}	Native acquisition-row index; it is not a physical r coordinate or an aligned or conditioned slot.
$j \in \{0, \ldots, N-1\}$	Ordered aligned input slot on the admitted ALIGN grid.
$n \in \{0, \ldots, N_y-1\}$	Ordered phase-zero RTC output slot.
$\mathcal{R}$	RTC application context naming requested use, paired input and interval, consumers, and constraints. Pointing, OOF, Science, Beammap, and diagnostic labels may be context facts; no such label restricts operation availability or silently selects policy.
$x^{\text{acq}}_{dm}, r^{\text{acq}}_{dm}$	Exactly paired raw readout coordinates on the native acquisition grid, before the imported ALIGN mapping.
x^{A}_{dj}, r^{A}_{dj}	Exactly paired admitted aligned raw KID readout coordinates and the RTC-local input boundary. x supplies the ordinary total-intensity analysis path; r supplies readout/leakage/artifact/Tune-health evidence. They share occurrence identity, cadence, phase, and mapping lineage but retain independent value validity.
$\mathbf{v}^{IQ,\text{acq}}_{dm}$	Upstream native-grid orthogonal microwave transmission components $(I^{\text{acq}}_{dm}, Q^{\text{acq}}_{dm})^{\mathsf{T}}$.
$\mathbf{u}^{xr}_{dj}$	Paired raw readout vector $(x^{A}_{dj}, r^{A}_{dj})^{\mathsf{T}}$ admitted at the RTC boundary.
ζ	Complete Tune and $I, Q \to x, r$ mapping identity, reference, normalization, validity domain, epoch, and uncertainty state.
χ_{dj}	Detector-static `flxscale` comparison coordinate used only to define compatible donor transfer; it is not a calibrated RTC product.
u^{x}_{dj}	Raw x-coordinate $\Delta f/f$ value immediately before the ordered RTC filter chain.
$v^{x,\text{preD}}_{dj}$	Final raw x stream after every admitted RTC replacement/filter stage and immediately before phase-zero point selection.
$y^{x,\text{RTC}}_{dn}$	Required conditioned raw x-coordinate $\Delta f/f$ RTC output at the selected phase-zero slot.
c^{CAL}_{dn}	Distinct downstream SCI-CAL output, when authorized: top-of-atmosphere, point-source-equivalent mJy per fixed nominal beam.
ϕ_d	`flxscale` for one exact detector occurrence under the declared multiplicative convention and calibration domain.
$A_\alpha, \mathbf{a}_\alpha$	Imported, realized upstream ALIGN mapping and affine term with state α. They map the native acquired pair to the admitted aligned pair, remain in lineage, and are not reapplied by the RTC-local operator.
C^{CAL}_{κ}	Downstream SCI-CAL operator with state κ, including selected absolute `flxscale` and target-atmosphere correction; RTC does not apply it.
S_σ	Spike-candidate evidence selector used only for masking or robust downweighting during level-shift learning; it does not replace samples.
B^{x}_{ω}	Realized post-segmentation x-replacement operator conditioned on selection state ω; it has no r numerical branch.
F_{k,θ_k}	Realized ordered FIR, IIR, notch, or admitted mask-controlled stage k.
$D_{M,0}$	Integer phase-zero point selector of factor $M \geq 1$.

Symbol	Normative meaning (continued)
Ω	Complete realized conditioning state, including all upstream bindings and every selected stage, exception, state, edge rule, and output interval.
$\mathcal{L}$	Declared learning population with exact identity, selection, support, masks, exclusions, and compatibility domain.
Λ	Learned evidence and uncertainty produced by declared diagnostics; it is not authorization to apply.
Π	One immutable resolved RTC plan admitted from learned evidence and fixed scientific policy.
$\mathcal{A}_\Pi$	Apply operator that consumes plan Π without changing it.
$a \in \{1, \ldots, A\}$	Refinement-attempt index; A is the number of attempts actually executed.
$A_{\max}$	Finite owner-approved maximum attempt count, with $A \leq A_{\max} < \infty$.
$k \in \{0, \ldots, K\}$	Accepted-plan index; $K \leq A$ is the final accepted-plan index.
k_a	Accepted-plan index available at the start of attempt a.
$\widetilde{\Pi}_a$	Complete cumulative successor plan proposed by attempt a; it is not an accepted plan unless its disposition is accepted.
δ_a	Typed attempt disposition in $\{\text{accepted}, \text{rejected}, \text{stop}\}$.
Π_k	Complete cumulative accepted plan k; Π_0 is the admitted initial fixed or identity plan.
$x^{\text{eval},(\text{k})}$	Exploratory evaluation product for accepted plan k, normally obtained by applying Π_k to original admitted input $x^{(0)}$.
ρ_{dn}	Representative aligned occurrence (d, Mn) on the raw paired ALIGN parent grid, before replacement and temporal propagation.
$\alpha_{x,d}, \alpha_{r,d}$	Local optical-response coefficients in the paired readout coordinates over a declared signal and support domain.
ϵ_d, ψ_d	Optional optical-leakage ratio $\alpha_{r,d}/\alpha_{x,d}$ and metric-qualified response-direction angle. Their coordinate units, scales, mapping revision, normalization, metric, domain, and comparison compatibility are part of the product; neither parameterization is universal.
$s^x_{d,\zeta}, s^r_{d,\zeta}$	Declared nonzero coordinate-normalization scales used only to place x/r response components in a common metric when an angle is reported.
$a_g(t)$	Declared atmospheric optical-mode template or latent common mode for network g.
$\tau_e, \delta\tau_{de}, \tau_{de}, \Delta\mathbf{b}_{de}$	Network event time, detector-specific timing offset, detector transition time $\tau_{de} = \tau_e + \delta\tau_{de}$, and paired amplitude $(\Delta x_{de}, \Delta r_{de})^\mathsf{T}$; the offset is zero or bounded by a declared coherence tolerance.
$I_h, \mathcal{P}_{d,h}$	Coherent plateau interval h and its application-context-specific admissibility state.
$K^{\text{RTC}}_{dn,qj}$	RTC-local detector-time response of one output cell to one admitted aligned input cell at fixed realized state.
$K^{\text{end}}_{dn,qm}$	Optional end-to-end response to one native acquired cell, available only by explicit composition with admitted upstream ALIGN response.
$\mathcal{S}_{dn}$	Exact transitive input support of output (d, n), including alignment, donors, filters, state, and selection.
$\mathcal{I}_{dn}$	Cause-preserving transitive influence set for output (d, n).

1.2 Identity and interval conventions

Observation, Tune, array, network/interface, detector occurrence, selected APT row/binding, native row, aligned slot, scan, science interval, context interval, conditioned output slot, stage, and product identities are non-interchangeable. Array identity is one of `a1100`, `a1400`, and `a2000`; array ID, array index, network ID, detector index, and container position do not substitute for one another. Cross-artifact identity requires an explicit occurrence- and product-scoped binding. Equal row numbers do not establish identity.

Primary detector timestreams are samples on rows and detectors on columns unless a product explicitly declares another ordered shape. Every interval is half-open. Science windows, physical scans, processing chunks, filter context, output selections, and learned-plan scan sets retain separate identities. Native, aligned, and output indices never silently renumber one another.

Time is seconds on the admitted ALIGN-assigned grid. The approximately 8-ms rate relationships and the 0.5×, 2×, and 4× family are conditional software-grid facts, not proof of a detector integration event, absolute epoch, or astrometric correction. Coordinate-dependent Pointing, OOF, and Beammap controls use a declared AltAz tangent plane; Science controls use a declared equatorial J2000 TAN relation. RTC performs no implicit frame conversion.

1.3 Logical and availability notation

$\mathbb{I}[\cdot]$ denotes an indicator, $\mathbb{E}$ expectation, Cov covariance, and supp$(\cdot)$ operator support. The logical symbols $\wedge$, $\vee$, and $\neg$ have their usual meanings. A quantity decorated available is present with its required identity, domain, and validity; a quantity decorated unavailable is explicitly unavailable. Neither zero nor NaN is an availability encoding.

End of notation; shared definitions follow without a change of authority.

2 Normative definitions

SCI-RTC-DEF-001 – Admitted aligned paired stream.
An immutable observation-scoped pair of `xs` and `rs` arrays. Every admitted x has exactly one r partner with the same observation, Tune, array, network/interface, detector occurrence, native acquisition occurrence, aligned slot, cadence, phase, and mapping lineage. Missing a partner is malformed required input. The paired samples retain independent finiteness, origin/synthesis, support, validity, and uncertainty states.

SCI-RTC-DEF-002 – RTC application context.
The immutable request-side description $\mathcal{R}$ naming requested use, paired input and interval, raw-coordinate sign and reference/baseline, valid domain, named consumers, and external constraints. Historical labels such as Beammap, Pointing, OOF, Science, or diagnostic may be recorded as context but do not partition the admitted RTC operation classes or silently select a policy. Every context consumes paired raw `xs`/`rs`; only conditioned x is eligible for a later SCI-CAL handoff. A calibrated downstream detector signal is a distinct SCI-CAL product c^{CAL} in the top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention; it is not another meaning of `xs`.

SCI-RTC-DEF-003 – Exact detector occurrence.
An observation-local layered binding containing observation, Tune, network/interface, network-local tone or admitted row slot, selected APT identity/digest, and row association. A transient column number is only a locator.

SCI-RTC-DEF-004 – Downstream CAL handoff.
An immutable RTC-to-SCI-CAL boundary carrying the raw conditioned parent, detector binding, complete RTC plan/response/support, selected `flxscale` authority, target-atmosphere lineage, validity, uncertainty availability, and provenance required by SCI-CAL. Absolute `flxscale` and target atmosphere are applied only after RTC; RTC does not derive, apply, repair, or promote the calibrated product. Because target-atmosphere correction is sample dependent and follows temporal filtering, a handoff claim also requires complete response accounting over RTC support or a preregistered owner-approved noncommutation bound; output-time correction is not an automatic inverse of a filtered atmosphere operator.

SCI-RTC-DEF-005 – Compatible `flxscale` pair.
Two finite factors ϕ_q and ϕ_d that are valid for the exact donor and target detector occurrences under the same multiplicative calibration convention, quantity domain, reference plane, and applicable support, with $\phi_d \neq 0$. Compatibility is not established by matching row numbers or finite values alone.

SCI-RTC-DEF-006 – Selected conditioning policy.
A versioned, admitted specification selected within a resolved RTC plan from operation classes available to every application context. It covers spike-candidate detection and change-point masking/downweighting, donor eligibility, donor scoring and ties, post-segmentation x replacement and fallback, source protection, filter stages and order, exact coefficients and normalization, numeric precision, state/reset, boundaries/edges, non-finite handling, sampling mode, phase-zero factor, flags, influence aggregation, diagnostics, and failure behavior. Availability or admission does not mean enabled, selected, numerically resolved, or scientifically qualified. It is selected authority, not an implementation default.

SCI-RTC-DEF-007 – One-way context–plan–record lifecycle.
Three distinct objects are preserved in one-way order. The *RTC application context* preserves requested use, input,

consumers, and constraints. The *RTC resolved plan* is the immutable selected numerical plan admitted after every required predicate passes, including any learned evidence and observation binding. The *RTC realized record* states the exact operations, intervals, state, exceptions, outputs, and completion that actually occurred. More detailed requested, effective, observation-resolved, learned, resolved, and realized substates may be retained, but a later object or substate never rewrites an earlier one.

SCI-RTC-DEF-008 – Realized conditioning operator.
The chronological composition of paired admission and evidence selection, level-shift transition masking and plateau segmentation, then every admitted x-domain replacement, mask, FIR/IIR/notch, boundary, reset state, non-finite, and phase-zero sampling action, including all affine state terms and exception intervals. Raw r may affect a selector or resolved state but does not thereby receive the x numerical operator. Annotations that affect values, support, influence, or availability are part of the operator.

SCI-RTC-DEF-009 – Phase-zero sampling.
Point selection of the pre-sampling x stream at input indices $j = Mn$ for one integer factor $M \geq 1$. It is not an arithmetic mean, block aggregate, or implicit anti-alias filter. Its representative occurrence is exactly (d, Mn) on the raw paired aligned parent before replacement and filtering; that identity is not a largest-coefficient claim.

SCI-RTC-DEF-010 – Complete realized response.
An exact local or factorized representation that includes every enabled response-changing stage, detector mixing, mask, state, edge rule, exception, and sampling phase on its declared support. The RTC-local response is with respect to the admitted aligned pair. An optional end-to-end response to the native acquisition grid is a distinct composition with admitted upstream ALIGN response and lineage. If either representation is not truthfully available, its response state is unavailable; a partial kernel is not complete response.

SCI-RTC-DEF-011 – Representative source occurrence.
For output (d, n), the raw aligned paired occurrence (d, Mn) before replacement and temporal propagation. It may be original, ALIGN-synthesized, or subsequently RTC-replaced. This identity, rather than the largest realized coefficient, governs independent-exposure exclusion.

SCI-RTC-DEF-012 – Transitive influence.
The cause-preserving closure from every original, synthesized, replaced, masked, edge, state, and non-finite source cell through all nonzero realized alignment, donor, filter, state, and sampling dependencies to an output. Influence is broader than the selected point and may cross detector columns.

SCI-RTC-DEF-013 – Direct exclusion and consumer eligibility.
An output whose representative occurrence of SCI-RTC-DEF-011 is ALIGN-synthesized or RTC-replaced is universally excluded as an independent detector measurement, even when finite. Synthesis or replacement elsewhere in its transitive support remains a mandatory cause-preserving influence input, not a universal RTC eligibility decision. Each downstream consumer owns the declared policy that maps that influence, response, and support to its own eligibility.

SCI-RTC-DEF-014 – Coordinate-dependent control.
A source mask or response operation that consumes an AST coordinate identity, frame, topology, detector binding, validity, uncertainty where required, and support. Invalid or unavailable coordinates are not outside-source values.

SCI-RTC-DEF-015 – Fixed sampling mode.
A request whose admitted factor, realizable prefilter, rate, phase-zero rule, and downstream compatibility are resolved without learning them from the observation.

SCI-RTC-DEF-016 – Learned sampling mode.
An optional request that evaluates an admitted finite candidate set and resolves one immutable observation-wide cadence/filter plan before apply. Its admission uses the smallest admitted beam, maximum valid in-scan speed, analytical beam-times-realized-filter response, exact phase-zero alias accounting, sampling sufficiency, filter realizability, and downstream compatibility. Noise-aware, per-array, per-scan, heterogeneous-cadence, and continuous adaptation are outside v0.1.

SCI-RTC-DEF-017 – Conditional statistical covariance.
The covariance obtained from an admitted input covariance through a fixed realized operator while holding selection and nuisance state fixed. It is not total uncertainty and does not imply a diagonal, white, stationary, or independent noise model. Every covariance or uncertainty claim, whether conditional, component-limited, or total, explicitly names the components and correlations it includes and excludes; unknown items remain unavailable rather than zero.

SCI-RTC-DEF-018 – Atomic RTC bundle.
One consumer-neutral completion unit containing conditioned raw x TOD; the immutable paired `rs` parent and pairing/mapping identity; every `rs`-derived diagnostic, selector, segmentation, and resolved-plan use; ordered detector/sample/scan/support and parent identity; complete response or typed unavailability; cause-preserving

flags and influence; separated validity and eligibility inputs; conditional uncertainty/covariance status; one-way state and provenance; required common diagnostics and any separately requested diagnostic detail; downstream CAL handoff state; and completion/failure state. A separately conditioned r product is optional and unavailable without its own declared operator, response, support, unit, validity, and provenance. TOD alone is not this bundle.

SCI-RTC-DEF-019 – Scientifically named diagnostic.
A quantity with an explicit estimator, unit, support, selection/mask, validity, uncertainty/availability, and policy role. It is either inert, advisory, or an input to an explicitly selected policy; those roles are not interchangeable.

SCI-RTC-DEF-020 – Claim layer.
One of algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, or production readiness. Evidence at one layer does not establish another.

SCI-RTC-DEF-021 – Declared learning population.
The exact training interval, designated same-observation interval, scan collection, observation aggregate, or other named population used to infer an RTC quantity, including identity, support, selection, masks, exclusions, signal role, compatibility domain, and completeness. A population is not defined by whatever samples happen to be available during execution.

SCI-RTC-DEF-022 – Learned evidence.
The output of named diagnostics applied to a declared learning population, including candidate parameters, estimator, unit and convention, support, quality, uncertainty, validity, and failure state. Learned evidence proposes or constrains a plan; it does not authorize apply.

SCI-RTC-DEF-023 – Resolved RTC plan.
One immutable observation- and application-context-bound specification admitted only after all required scientific predicates pass. It contains operation sequence, family, exact coefficients or identity, rates and conventions, order/taps, bands and tolerances, notch state, applicability, masks, donor policy, edges, guards, resets, decimation and timing, expected response/support, uncertainty, validity, and downstream compatibility.

SCI-RTC-DEF-024 – Apply state.
Execution of exactly one resolved plan on compatible admitted input. Apply records actual intervals, exceptions, resets, outputs, and completion but does not relearn, retune, change policy, or rewrite the resolved plan.

SCI-RTC-DEF-025 – Online-adaptive estimator.
A separate time-varying, data-dependent estimator whose coefficients or other operator parameters evolve during apply under a declared update rule. Its learning support, stability, response, uncertainty, guard, reset, and restart semantics are not those of learned-then-fixed filtering. It is unavailable in v0.1 absent successor authority.

SCI-RTC-DEF-026 – Scientific filter design specification.
The physical purpose, source/contaminant model, admitted scan/beam/source domain, pass/transition/stop or notch constraints, phase and registration, alias bound, support/guard, numeric precision, realizability, validation, and failure criteria that justify a filter family and its numerical parameters. An operation label or coefficient array alone is not this specification.

SCI-RTC-DEF-027 – Bounded iterative notch-plan refinement.
A finite owner-bounded sequence of refinement attempts. Each attempt learns from a declared population and the current accepted-plan evaluation product, resolves a complete cumulative proposal and typed disposition, and advances the accepted-plan index only when that proposal is accepted. Apply of every accepted plan is immutable; rejected or stop attempts do not manufacture a new accepted plan or evaluation product. This is not the online-adaptive estimator of SCI-RTC-DEF-025.

SCI-RTC-DEF-028 – Complete cumulative successor proposal.
The full plan proposed by attempt a and accepted as Π_{k+1} only after its admission predicates pass, containing every retained, modified, removed, and rejected notch disposition; every broad-band stage; exact order, coefficients, precision, state, masks, edges, guards, intervals, decimation, timing, cumulative response/support, uncertainty, validity, and downstream compatibility. "Add this notch to the prior output" is not a complete successor plan.

SCI-RTC-DEF-029 – Final accepted plan and termination disposition.
The last complete plan whose admission predicates passed, together with one typed finite stop cause. No-successor, residual-satisfied, stable-plan, budget/stability/support/compatibility rejection, insufficient evidence, maximum-attempt, oscillating-plan, nonconvergent-parameter, and owner-cancelled states remain distinct. Maximum-attempt termination is not convergence.

SCI-RTC-DEF-030 – Paired raw x, r detector coordinates.
The two raw coordinates $\mathbf{u}_d = (x_d, r_d)^\mathsf{T}$ produced for one exact detector occurrence by an admitted Tune-dependent readout mapping. The x coordinate carries the ordinary total-intensity analysis path; the nominally orthogonal r coordinate carries readout, leakage, artifact, and Tune-health information. Pairing does not assert equal scaling, response, validity, or uncertainty.

SCI-RTC-DEF-031 – Tune-dependent $I, Q \to x, r$ mapping authority.
The upstream authority identified by ζ that owns the microwave transmission coordinates, transform, reference point, normalization, sign, local Jacobian or nonlinear equivalent, validity/linearity domain, epoch, and uncertainty. RTC consumes its identity and admitted output but does not produce or repair the mapping.

SCI-RTC-DEF-032 – Optical leakage into the paired r coordinate.
A nonzero admitted optical-response projection in nominally orthogonal r. It is a readout-coordinate alignment or optical-leakage diagnostic, not automatically a literal Tune-angle error. A scalar ratio or angle tests only its declared optical projection and does not prove the complete transform. Every product retains the x/r coordinate units or scales, mapping revision, normalization and metric, Tune and validity domain, whether its ratio is dimensionless or carries a coordinate-unit ratio, and the compatibility rule for comparison across detectors, Tunes, loading states, epochs, or mapping revisions. An angle is available only after a declared metric makes the two axes commensurate. Otherwise it is coordinate bookkeeping, not a comparable physical readout angle.

SCI-RTC-DEF-033 – Atmospheric optical-leakage diagnostic.
A scientifically named estimate of paired x/r response to a declared network atmospheric optical-mode template, with exact membership, self-inclusion rule, time/frequency support, preprocessing, estimator, response, covariance, uncertainty, coherence, Tune/loading state, and scalar, frequency-dependent, or unavailable disposition. The estimator shall address shared-data dependence, noise in both coordinates, and detector self-inclusion; measured correlation or direct noisy r-on-x regression is not by itself an unbiased leakage coefficient. This diagnostic does not authorize numerical template subtraction from x.

SCI-RTC-DEF-034 – Bright-source optical-leakage diagnostic.
A separately parented estimate of paired x/r response to a declared localized source template and support, often requested by Beammap, Pointing, or OOF application contexts. It neither inherits nor silently replaces an atmosphere-derived diagnostic.

SCI-RTC-DEF-035 – Network-coherent detector-dependent level shift.
An unexpected persistent additive readout-baseline change having a common or near-common network event identity, detector transition support constrained by a declared coherence tolerance, and independently estimated detector/channel amplitudes $(\Delta x, \Delta r)$. On each stable x plateau the v0.1 model is signal plus an additive baseline; it contains no multiplicative gain, responsivity, or more general response-change degree of freedom. Simultaneity, an r signature, or its absence does not by itself identify physical cause.

SCI-RTC-DEF-036 – Transition interval and guarded change-point support.
The finite physical-time interval between stable pre- and post-event plateaus, together with its timing uncertainty, affected detector population, and any downstream operator-specific guards. Its sample support is derived from the applicable timing vector, never from a fixed sample count or scan-relative fraction. The physical transition support is explicitly flagged, excluded from both plateau estimators, and remains unmodeled even when a valid plateau offset correction is applied. Propagated influence remains distinguishable from this underlying physical support. Reset is the ordinary segmentation behavior; carry requires separately authorized continuity.

SCI-RTC-DEF-037 – Stable plateau and application-context admissibility.
A half-open interval outside resolved transition support that is consistent with its declared signal-plus-additive-baseline model, accounting for atmosphere, sources, noise, slow variation, Tune state, valid support, and absence of unresolved additional shifts. Suitability is application-context specific and depends on the required estimator support, not sample count alone.

SCI-RTC-DEF-038 – Separately authorized response-changing correction.
Any estimator that numerically mixes x and r, subtracts or rotates one into the other, or assigns a multiplicative gain, responsivity, or more general response change to a level-shift event. It requires separate purpose, coupling and learning models, complete response and covariance, astronomical transfer, uncertainty, validation, restart, and provenance authority; none is part of the v0.1 additive level-shift model. A selected additive translation between sufficiently supported stable x plateaus is governed instead by Equations (SCI-RTC-EQ-033)–(SCI-RTC-EQ-034) and REQ-101.

3 Normative equations and identities

The equations in this section are the sole displayed mathematical authority of both views. They describe an implementation-independent, state-conditioned operator. They neither select numerical policy nor assert implementation conformity.

3.1 Raw RTC domain, donor transfer, and CAL handoff

For an exact detector occurrence i under the admitted multiplicative `flxscale` convention, define the detector-static comparison coordinate

$$\chi_i = \phi_i x_i, \qquad \phi_i = \texttt{flxscale}_i. \tag{SCI-RTC-EQ-001}$$

Equation (SCI-RTC-EQ-001) neither produces a calibrated RTC product nor assigns a role to legacy `responsivity`; target-atmosphere correction is absent. When q is a raw donor and d is a raw target, the unique multiplicative transfer that represents donor-calibrated amplitude in the target raw convention is

$$x^R_{d\leftarrow q} = a_{d\leftarrow q} x_q, \qquad a_{d\leftarrow q} = \frac{\phi_q}{\phi_d}, \qquad \chi^R_d = \phi_d x^R_{d\leftarrow q} = \phi_q x_q = \chi_q. \tag{SCI-RTC-EQ-002}$$

This equation is defined only for a compatible `flxscale` pair as defined above. Otherwise this transfer is unavailable. An additive calibration convention, a different unit/reference domain, or an alternative donor authority requires its own complete affine equivalence and is not silently folded into equation (SCI-RTC-EQ-002).

Let $\Phi = \mathrm{diag}(\phi_d)$ be the selected detector-static factor operator and $C^{\mathrm{atm}}_{\kappa,\mathcal{R}}$ the target-atmosphere operator owned by SCI-CAL. The distinct downstream calibration operator is

$$C^{\mathrm{CAL}}_{\kappa,\mathcal{R}} = \Phi C^{\mathrm{atm}}_{\kappa,\mathcal{R}}, \qquad \mathbf{c}^{\mathrm{CAL}}_{\mathcal{R}} = C^{\mathrm{CAL}}_{\kappa,\mathcal{R}} \mathbf{y}^{x,\mathrm{RTC}}_{\mathcal{R}}. \tag{SCI-RTC-EQ-003}$$

Equation (SCI-RTC-EQ-003) is an SCI-CAL handoff identity; its output is not `xs` and it is not an RTC stage. The selected cross-package boundary is

$$\begin{aligned}
\begin{bmatrix}\mathbf{x}^A\\ \mathbf{r}^A\end{bmatrix} &= A_\alpha \begin{bmatrix}\mathbf{x}^{\mathrm{acq}}\\ \mathbf{r}^{\mathrm{acq}}\end{bmatrix} + \mathbf{a}_\alpha && \text{(upstream ALIGN relation)},\\
(\mathbf{x}^A, \mathbf{r}^A) &\xrightarrow{\text{RTC pair admission, spike evidence, shift resolve}} \Omega_{xr},\\
\mathbf{x}^A &\xrightarrow{B^x_\omega,\, F_1\ldots F_K,\, D_{M,0}} \mathbf{y}^{x,\mathrm{RTC}}_{\mathcal{R}} \xrightarrow{C^{\mathrm{CAL}}_{\kappa,\mathcal{R}}} \mathbf{c}^{\mathrm{CAL}}_{\mathcal{R}},\\
\mathbf{r}^A &\xrightarrow{\text{retain parent and declared diagnostics}} \mathcal{D}^r_{\mathcal{R}} \hookrightarrow \mathcal{O}_{\mathrm{RTC},\mathcal{R}}.
\end{aligned} \tag{SCI-RTC-EQ-004}$$

RTC shall not move the downstream CAL arrow inside replacement, temporal filtering, or phase-zero sampling. Only the x member is handed to SCI-CAL. The paired raw r parent and every causal diagnostic/selector remain in the bundle but are not calibrated, substituted for x, or interpreted as Stokes data. The raw donor ratio in equation (SCI-RTC-EQ-002) is x-domain convention transfer within B^x_ω, not absolute calibration. Atmospheric templates are diagnostic evidence only; equation (SCI-RTC-EQ-004) contains no atmospheric-removal operator. The first line is retained upstream lineage, not an RTC stage. RTC begins at the admitted aligned pair; A_α is composed again only when a separately identified end-to-end response to native acquisition coordinates is requested.

3.2 Realized conditioning operator

Conditioned on complete realized state Ω—which may depend on paired x/r evidence and selectors—the required ordered raw x operator is

$$\boxed{\mathbf{y}^{x,\mathrm{RTC}}_{\mathcal{R}} = D_{M,0} F_{K,\theta_K} \cdots F_{1,\theta_1} B^x_{\omega,\mathcal{R}} \mathbf{x}^A + \mathbf{b}^x_\Omega\ .} \tag{SCI-RTC-EQ-005}$$

Chronological order is read from right to left. The affine terms include only declared replacement, filter-state, boundary, or other authorized contributions. Masks, coefficients, finite handling, context, exceptions, and state are components of Ω, not annotations outside the operator. Conditional on them,

$$\begin{aligned}
\mathbf{y}^{x,\mathrm{RTC}}_{\mathcal{R}} &= L^x_\Omega \mathbf{x}^A + \mathbf{b}^x_\Omega, && L^x_\Omega = D_{M,0} F_K \cdots F_1 B^x_{\omega,\mathcal{R}},\\
K^{x\leftarrow x,\mathrm{RTC}}_\Omega &= \left.\frac{\partial \mathbf{y}^x}{\partial \mathbf{x}^A}\right|_\Omega = L^x_\Omega, \quad K^{x\leftarrow r,\mathrm{RTC}}_\Omega = \left.\frac{\partial \mathbf{y}^x}{\partial \mathbf{r}^A}\right|_\Omega = 0,\\
J^{\mathrm{end}}_\Omega &= \begin{bmatrix} L^x_\Omega & 0\end{bmatrix} A_\alpha && \text{when the upstream ALIGN response is admitted.}
\end{aligned} \tag{SCI-RTC-EQ-006}$$

This is a conditional affine identity, not an unconditional linear, stationary, unbiased, or Gaussian claim.

A realized replacement at target cell (d,j) may be written

$$x^R_{dj} = (1-p_{dj})x_{dj} + p_{dj}\left[\sum_{(q,\ell)\in\mathcal{Q}_{dj}} w_{dj,q\ell}\, a_{d\leftarrow q,j\ell}\, x_{q\ell} + \beta_{dj}\right], \quad \sum_{q,\ell} w_{dj,q\ell} = 1 \text{ when normalization is claimed.} \tag{SCI-RTC-EQ-007}$$

This replacement is explicitly x-only. The scale a is domain-qualified: equation (SCI-RTC-EQ-002) is the authorized raw `flxscale` route. Any other transfer carries separate identity and validity and does not import downstream calibration into RTC.

For a time-invariant FIR stage,

$$v_{d,j} = \sum_{\ell\in\mathcal{L}_k} h^{(k)}_{d,\ell} u_{d,j-\ell}, \qquad H^{(k)}_d(e^{i\omega}) = \sum_{\ell\in\mathcal{L}_k} h^{(k)}_{d,\ell} e^{-i\omega\ell}, \qquad H^{(k)}_d(1) = \sum_{\ell} h^{(k)}_{d,\ell}. \tag{SCI-RTC-EQ-008}$$

For a causal IIR or notch stage, a state-complete form is

$$\mathbf{s}_{j+1} = A_k\mathbf{s}_j + B_k u_j, \qquad v_j = C_k\mathbf{s}_j + D_k u_j, \qquad \mathbf{s}_{j_0} = \mathbf{s}_{0,k}. \tag{SCI-RTC-EQ-009}$$

An exactly equivalent recurrence is admissible only with its coefficient convention, precision, state, and boundary identity. A causal-stability claim requires every realized pole strictly inside the unit circle.

One possible selected mask operator is normalized masked FIR convolution,

$$v_j = \frac{\sum_\ell h_\ell m_{j-\ell} u_{j-\ell}}{\sum_\ell h_\ell m_{j-\ell}}, \qquad \left|\sum_\ell h_\ell m_{j-\ell}\right| > \epsilon_m, \tag{SCI-RTC-EQ-010}$$

but equation (SCI-RTC-EQ-010) is not a selected default. Segmenting, zero filling, restoring a mask after filtering, holding state, and renormalizing are distinct operators.

3.3 Phase-zero sampling and response

For a coherent pre-sampling segment of length N, v0.1 selects only

$$y^{x,\mathrm{RTC}}_{d,n} = v^{x,\mathrm{preD}}_{d,Mn}, \qquad 0 \le Mn < N, \qquad f_{\mathrm{out}} = \frac{f_{\mathrm{in}}}{M}, \qquad N_y = \begin{cases} 0, & N = 0, \\ 1 + \left\lfloor \dfrac{N-1}{M} \right\rfloor, & N > 0. \end{cases} \tag{SCI-RTC-EQ-011}$$

The representative assigned-grid time is the selected point time $t^{\mathrm{out}}_n = t_{Mn}$; the scientific support is nevertheless the complete RTC-local transitive support $\mathcal{S}^{\mathrm{RTC}}_{dn}$ of all preceding stages. A selected edge/output policy may reject candidate points, but shall not relabel a different phase or silently change equation (SCI-RTC-EQ-011). The representative source occurrence is exactly $\rho_{dn} = (d, Mn)$ in the raw aligned paired parent. It is not the largest response coefficient, a support centroid, a donor occurrence, or a synthetic sample.

For output (d,n) and admitted aligned input (q,j), the exact RTC-local response is

$$K^{\mathrm{RTC}}_{dn,qj} = \frac{\partial y_{dn}}{\partial x^A_{qj}} \quad \text{at fixed realized state } \Omega. \tag{SCI-RTC-EQ-012}$$

At a selection boundary the derivative may be discontinuous or undefined; that state shall be represented rather than replaced by an LTI claim. An end-to-end kernel to native acquisition cells is the distinct optional composition already defined in equation (SCI-RTC-EQ-006).

In the fixed, detector-separable, translation-invariant, single-rate interior case only,

$$H^{\mathrm{comb}}_d(e^{i\omega}) = \prod_{k=1}^{K} H^{(k)}_d(e^{i\omega}), \qquad \tau_g(\omega) = -\Delta t\,\frac{d}{d\omega}\,\mathrm{unwrap}\big[\arg H^{\mathrm{comb}}_d(e^{i\omega})\big]. \tag{SCI-RTC-EQ-013}$$

Amplitude is $|H|$, power response is $|H|^2$, and phase is $\arg H$ under the recorded Fourier convention. Group delay is unavailable at response zeros unless a bounded convention is supplied.

For an LTI prefilter followed by phase-zero selection, the exact multirate identity is

$$Y(e^{i\Omega_o}) = \frac{1}{M}\sum_{p=0}^{M-1} H\left(e^{i(\Omega_o+2\pi p)/M}\right) X\left(e^{i(\Omega_o+2\pi p)/M}\right), \tag{SCI-RTC-EQ-014}$$

under the declared Fourier/sign convention. Every folded image contributes; point selection itself is not anti-alias filtering.

For an FIR output, direct support starts with

$$\mathcal{S}^{(1)}_{dn} = \{Mn - \ell : \ell \in \mathcal{L}\}, \qquad \mathcal{S}^{\mathrm{RTC}}_{dn} = \mathrm{closure}_{B,F,\,\mathrm{state}}\left(\mathcal{S}^{(1)}_{dn}\right). \tag{SCI-RTC-EQ-015}$$

This is RTC-local support on the aligned grid. Optional end-to-end support is its explicit upstream expansion through admitted A_α support. IIR support extends to the declared state/reset or context boundary and may be mathematically infinite. Any finite guard is an approximation with a stated residual-tail criterion.

3.4 Moments, influence, and learned-plan selection

For admitted paired input $\mathbf{u}^A = (\mathbf{x}^A, \mathbf{r}^A)^\mathsf{T}$ with mean $\boldsymbol{\mu}_u$ and joint covariance $\Sigma_u = \left[\begin{smallmatrix}\Sigma_{xx} & \Sigma_{xr}\\ \Sigma_{rx} & \Sigma_{rr}\end{smallmatrix}\right]$, conditional on fixed Ω independent of the modeled random noise, define $J_{\mathrm{num},\Omega} = \begin{bmatrix} L^x_\Omega & 0\end{bmatrix}$. Then

$$\mathbb{E}[\mathbf{y} \mid \Omega] = J_{\mathrm{num},\Omega}\boldsymbol{\mu}_u + \mathbf{b}^x_\Omega, \tag{SCI-RTC-EQ-016a}$$

$$\Sigma^{\mathrm{stat}}_{y|\Omega} = J_{\mathrm{num},\Omega}\Sigma_u J^\mathsf{T}_{\mathrm{num},\Omega} = L^x_\Omega \Sigma_{xx} (L^x_\Omega)^\mathsf{T}. \tag{SCI-RTC-EQ-016b}$$

The cross-coordinate blocks do not enter fixed-state numerical propagation. They remain required for a joint selector, estimator, learned parameter, or diagnostic because $\Omega = \Omega(\mathbf{x}^A, \mathbf{r}^A)$. If selectors depend on the random data, unconditional covariance obeys

$$\mathrm{Cov}(\mathbf{y}) = \mathbb{E}_\Omega[\mathrm{Cov}(\mathbf{y} \mid \Omega)] + \mathrm{Cov}_\Omega(\mathbb{E}[\mathbf{y} \mid \Omega]). \tag{SCI-RTC-EQ-017}$$

The second term is not zero by default. With named nuisance parameters $\boldsymbol{\eta}$ and an adequate local linearization,

$$\Sigma^{\mathrm{sys}}_y \approx J_\eta \Sigma_\eta J^\mathsf{T}_\eta, \qquad J_\eta = \left.\frac{\partial \mathbb{E}[\mathbf{y} \mid \Omega, \boldsymbol{\eta}]}{\partial \boldsymbol{\eta}}\right|_{\boldsymbol{\eta}_0}, \tag{SCI-RTC-EQ-018}$$

Claims may be conditional, component-limited, or total. Each claim shall name the components and correlations included and excluded. The following is a possible complete decomposition only when every applicable term and cross term is supplied; "total" is reserved for that declared complete coverage:

$$\Sigma^{\mathrm{total}}_y = \Sigma^{\mathrm{stat}}_y + \Sigma^{\mathrm{sys}}_y + \Sigma^{\mathrm{selection}}_y + \Sigma^{\mathrm{model}}_y + \Sigma^{\mathrm{cross}}_y. \tag{SCI-RTC-EQ-019}$$

Unknown or omitted components and correlations are unavailable, not zero. A truthfully labeled partial or conditional claim does not require this maximal decomposition.

Define cause-preserving influence, direct exclusion, and consumer eligibility by

$$\mathcal{I}_{dn} = \mathrm{closure}_{K,\,\mathrm{generative\ links}}\{\text{typed source causes on } \mathcal{S}_{dn}\}, \tag{SCI-RTC-EQ-020a}$$

$$E^{\mathrm{RTC,direct}}_{dn} = V^{\mathrm{det}}_d \wedge V^{\mathrm{num}}_{dn} \wedge V^{\mathrm{support}}_{dn} \wedge V^{\mathrm{response}}_{dn} \wedge \neg D^{\mathrm{ALIGN\_synth}}_{dn} \wedge \neg D^{\mathrm{RTC\_replace}}_{dn},$$

$$E^{(c)}_{dn} = E^{\mathrm{RTC,direct}}_{dn} \wedge P_c(\mathcal{I}_{dn}, \mathcal{S}_{dn}, K_{dn,\cdot\cdot}) \tag{SCI-RTC-EQ-020b}$$

D_{dn} identifies whether the exact representative occurrence ρ_{dn} was synthesized or replaced before subsequent filtering and selection. P_c is the declared policy of consumer c for noncenter transitive influence. RTC preserves every cause and numerical dependency but does not replace P_c with a universal transitive-ineligibility decision.

For learned mode, let $\mathcal{M}$ be the admitted finite integer candidate set and $\mathcal{F}(M)$ the conjunction of all owner-resolved astronomical transfer, exact phase-zero alias, sampling, realized-filter realizability, and downstream-compatibility constraints, evaluated with the smallest admitted beam and maximum valid in-scan speed. Then

$$\mathcal{M}_{\mathrm{safe}} = \{M \in \mathcal{M} : \mathcal{F}(M) = \mathrm{true}\}, \qquad M_\star = \max \mathcal{M}_{\mathrm{safe}}. \tag{SCI-RTC-EQ-021}$$

Equation (SCI-RTC-EQ-021) is defined only after every numerical tolerance, support rule, fallback, and compatibility predicate named in the owner ledger is resolved. Percentile speed is diagnostic, not safety authority.

Finally, the consumer-neutral atomic product for application context $\mathcal{R}$ is

$$\mathcal{O}_{\mathrm{RTC},\mathcal{R}} = (\mathbf{y}^x, \mathbf{r}^{A,\mathrm{parent}}, \mathcal{J}^{xr}, \mathcal{K}^x, \mathcal{S}, \mathcal{I}, \mathcal{F}, \mathcal{V}, \mathcal{U}, \mathcal{P}, \mathcal{D}, \mathcal{C}), \tag{SCI-RTC-EQ-022}$$

where $\mathcal{J}^{xr}$ is exact ordered pair identity, $\mathcal{K}^x$ is the complete conditioned-x response status, $\mathbf{r}^{A,\mathrm{parent}}$ is the immutable paired raw parent rather than an implied conditioned output, $\mathcal{S}$ support, $\mathcal{I}$ influence, $\mathcal{F}$ typed flags/causes, $\mathcal{V}$ separated validity and eligibility inputs, $\mathcal{U}$ uncertainty/covariance availability, $\mathcal{P}$ one-way state/provenance, $\mathcal{D}$ diagnostics, and $\mathcal{C}$ atomic completion or required-output failure state.

3.5 Scientific design and learn–resolve–apply identities

For a locally Gaussian effective beam with angular covariance Σ and angular scan-velocity vector $\mathbf{v}$, the projected source-crossing time scale is

$$\sigma_t = \left(\mathbf{v}^{\mathsf{T}}\Sigma^{-1}\mathbf{v}\right)^{-1/2}. \tag{SCI-RTC-EQ-023}$$

Equation (SCI-RTC-EQ-023) is a local Gaussian diagnostic. Scientific admission shall use the exact source-times-realized-response model over the declared beam, scan, source-class, and cadence domain.

A generic realized notch may be represented, where applicable, by

$$H_{\mathrm{notch}}(z) = g\,\frac{\prod_{a=1}^{N_z}(1 - z_a z^{-1})}{\prod_{b=1}^{N_p}(1 - p_b z^{-1})}, \qquad |p_b| < 1 \text{ for a causal-stability claim.} \tag{SCI-RTC-EQ-024}$$

This form imposes no universal biquad. Exact realized coefficients, sample rate, frequency convention, phase/direction, state, edges, and precision are part of the response.

For an admitted FIR design family H_N, a selected order shall follow a declared constrained problem of the form

$$N_\star = \min_N \left\{ N : \begin{array}{l} \sup_{f\in\mathcal{P}} |H_N(f) - H_{\mathrm{target}}(f)| \le \epsilon_{\mathcal{P}}, \\ \sup_{f\in\mathcal{S}} |H_N(f)| \le \epsilon_{\mathcal{S}}, \\ \text{phase/delay, alias, edge/guard, precision, and realizability predicates pass} \end{array} \right\}. \tag{SCI-RTC-EQ-025}$$

The sets, metrics, tolerances, equality conventions, and family are owner-resolved; an approximate order formula is not universal authority.

For a causal linear-phase FIR with N taps at rate f_{samp}, its nominal interior group delay is

$$\tau_g = \frac{N-1}{2f_{\mathrm{samp}}}. \tag{SCI-RTC-EQ-026}$$

Where a temporal response has effective delay τ during a locally uniform scan, the corresponding scan-direction angular displacement is

$$\Delta\boldsymbol{\theta} = \mathbf{v}\,\tau. \tag{SCI-RTC-EQ-027}$$

For every learned RTC operation, the lifecycle is

$$\Lambda = \mathrm{Learn}(\mathcal{L};\Theta^{\mathrm{eff}}), \qquad \Pi = \mathrm{Resolve}(\Lambda, \Theta^{\mathrm{policy}}), \qquad \mathbf{y} = \mathcal{A}_\Pi(\mathbf{x}), \qquad \frac{\partial\Pi}{\partial j} = 0 \text{ during apply.} \tag{SCI-RTC-EQ-028}$$

The final equality denotes plan immutability over apply slots; it is not an assumption that the realized operator is stationary at masks, edges, state boundaries, or other declared exceptions.

For bounded iterative notch-plan refinement with original admitted input $x^{(0)}$, initial fixed or identity plan Π_0, actual attempt count A, configured maximum $A_{\max}$, attempt index $a \in \{1,\ldots,A\}$, and accepted-plan index k_a available at the start of attempt a,

$$
\begin{aligned}
x^{\mathrm{eval},(0)} &= \mathcal{A}_{\Pi_0}(x^{(0)}), \qquad k_1 = 0, \\
\Lambda_a &= \mathrm{Learn}(\mathcal{L}_a, x^{\mathrm{eval},(\mathrm{k_a})}, \Pi_{k_a}), \\
(\widetilde{\Pi}_a, \delta_a) &= \mathrm{Resolve}(\Lambda_a, \Pi_{k_a}, \Theta^{\mathrm{policy}}), \\
k_{a+1} &= k_a + \mathbb{I}[\delta_a = \mathrm{accepted}], \\
\delta_a = \mathrm{accepted} \implies & \begin{cases} \Pi_{k_a+1} = \widetilde{\Pi}_a, \\ x^{\mathrm{eval},(\mathrm{k_a+1})} = \mathcal{A}_{\Pi_{k_a+1}}(x^{(0)}) \end{cases}, \\
\delta_a \neq \mathrm{accepted} \implies & \Pi_{k_{a+1}} = \Pi_{k_a} \quad \text{and no new evaluation is required}, \\
x^{\mathrm{final}} &= \mathcal{A}_{\Pi_K}(x^{(0)}), \qquad K = k_{A+1} \leq A \leq A_{\max} < \infty, \\
\frac{\partial \Pi_k}{\partial j} &= 0 \quad \text{during every apply of accepted plan } k.
\end{aligned}
\tag{SCI-RTC-EQ-029}
$$

Equation (SCI-RTC-EQ-029) separates the actual finite attempt count from its configured maximum and from the accepted-plan sequence. It does not create artificial no-op attempts after an earlier stop. It defines the initial evaluation for nonidentity Π_0, and states the default original-input replay rule. A sequential cascade from a preceding evaluation product requires separate explicit operator authority and complete propagation of every intermediate parent, response, state, phase, edge, support, covariance, and uncertainty.

3.6 Paired coordinates, leakage, and level shifts

For every native acquired detector occurrence, the upstream mapping authority shall produce one ordered pair from measured in-phase and quadrature coordinates before the separate ALIGN relation of equation (SCI-RTC-EQ-004),

$$
\begin{bmatrix} x^{\mathrm{acq}}_{dm} \\ r^{\mathrm{acq}}_{dm} \end{bmatrix} = \mathcal{T}_{d,\zeta}(I^{\mathrm{acq}}_{dm}, Q^{\mathrm{acq}}_{dm}), \qquad \mathcal{J}^{xr,\mathrm{acq}}_{dm} = (d, m, \mathrm{stream}, \zeta). \tag{SCI-RTC-EQ-030}
$$

The pair identity is exact. Missing either member is malformed required input; validity may nevertheless differ between the two members. An affine matrix, offset, or Jacobian is only a local representation over a declared domain of the generally nonlinear mapping $\mathcal{T}_{d,\zeta}$.

An admitted local optical response model may be written

$$
\begin{bmatrix} \Delta x_d \\ \Delta r_d \end{bmatrix} = \begin{bmatrix} \alpha^x_d \\ \alpha^r_d \end{bmatrix} s_d + \begin{bmatrix} \eta^x_d \\ \eta^r_d \end{bmatrix}, \qquad \alpha^x_d \neq 0. \tag{SCI-RTC-EQ-031}
$$

This is diagnostic authority, not permission to correct one member with the other or to assume $\alpha^r_d = 0$.

For a declared source class c and support-qualified estimator,

$$
\begin{aligned}
\epsilon^{(c)}_d &= \frac{\alpha^{r,(c)}_d}{\alpha^{x,(c)}_d}, \qquad && \Delta r_d = \epsilon^{(c)}_d \Delta x_d + \eta^{(c)}_d, \\
\psi^{(c)}_d &= \mathrm{atan2}\Big(\alpha^{r,(c)}_d / s^r_{d,\zeta}, \alpha^{x,(c)}_d / s^x_{d,\zeta}\Big) && \text{only under a declared commensurate metric.}
\end{aligned}
\tag{SCI-RTC-EQ-032}
$$

Atmosphere-derived and bright-source-derived estimates are distinct products with distinct parentage, support, validity, uncertainty, and scalar-versus- frequency-dependent status. The ratio carries the coordinate-unit ratio when x and r are not dimensionless under a common normalization. Comparison across mapping revisions is unavailable unless their units, scales, normalization, metric, and domains are compatible or an explicit transform has been applied.

For a declared level-shift event e, detector d has finite transition support $\mathcal{T}_{de}$ on the admitted aligned timing vector t^A_j. The v0.1 stable-plateau model is additive:

$$
\begin{gathered}
\mathcal{T}_{de} = [t^-_{de}, t^+_{de}], \qquad \Delta t_{de} = t^+_{de} - t^-_{de} > 0, \\
\tau_{de} \in \mathcal{T}_{de}, \qquad \tau_{de} = \tau_e + \delta\tau_{de}, \qquad |\delta\tau_{de}| \leq \Delta\tau^{\mathrm{coh}}_e, \\
\mathcal{I}_{de} = \{j : t^A_j \in \mathcal{T}_{de}\}, \\
x^A_{dj} = s^x_{dj} + b^-_{de} \quad (t^A_j < t^-_{de}), \qquad x^A_{dj} = s^x_{dj} + b^+_{de} \quad (t^A_j > t^+_{de}), \qquad \Delta b_{de} = b^+_{de} - b^-_{de}.
\end{gathered}
\tag{SCI-RTC-EQ-033}
$$

No plateau equation applies on $\mathcal{I}_{de}$. The network event identity, detector interval localization or conservative bound, physical-time width, timing-vector-derived sample support, coherence tolerance, separately measured x/r evidence or unavailability, and resulting segment boundaries belong to the realized record. No multiplicative gain or responsivity term is admitted.

For sufficiently supported stable pre/post plateau sets $\mathcal{G}^-_{de}, \mathcal{G}^+_{de}$, an admitted additive correction and reset condition may be represented by

$$\begin{aligned} &\widehat{b}^{\pm}_{de} = \mathrm{Plateau}(x^A_{d,\mathcal{G}^{\pm}_{de}}; \Theta_P), \qquad \widehat{\Delta b}_{de} = \widehat{b}^{+}_{de} - \widehat{b}^{-}_{de}, \qquad \mathcal{G}^{\pm}_{de} \cap \mathcal{I}_{de} = \varnothing, \\ &x^{A,\mathrm{corr}}_{dj} = x^A_{dj} - \widehat{\Delta b}_{de}\, \mathbb{I}[t^A_j > t^+_{de}] \quad (j \notin \mathcal{I}_{de}), \qquad \mathbf{s}_{j^+_g} = R_g(\mathbf{s}_{j^-_g}; \Theta_R), \end{aligned} \tag{SCI-RTC-EQ-034}$$

This displayed orientation translates the post-event plateau to the pre-event reference; a selected plan may use the opposite reference and sign. The transition cells remain flagged and unmodeled, not values made scientifically valid by the offset. If either required stable support or its selected quality/uncertainty criterion fails, $\widehat{\Delta b}_{de}$ and the correction are unavailable. R_g is ordinarily reset; carry across an accepted shift remains separately authorized continuity.

The actual bounded-refinement counts satisfy

$$0 \le K \le A \le A_{\max} < \infty, \qquad A = \#\{a : \mathrm{Learn}_a \ \text{ was actually attempted}\}. \tag{SCI-RTC-EQ-035}$$

4 Bounded assumptions and non-goals

SCI-RTC-ASM-001 – Capability.
V0.1 admits the exact paired raw readout coordinates `xs` and `rs`. The x member is the ordinary science-conditioning path; the r member is a retained diagnostic coordinate. Raw x is not itself a Stokes observable, and enabled polarimetry remains excluded until separately approved.

SCI-RTC-ASM-002 – Aligned-grid boundary.
The input grid, cadence, phase, mapping, gaps, synthesis/origin state, support, and scan partition are admitted ALIGN facts. Physical event timing, absolute timing accuracy, and a producer epoch may remain unavailable. RTC's local numerical operator begins at this aligned pair; ALIGN remains upstream lineage and is composed only for a separately identified end-to-end response.

SCI-RTC-ASM-003 – Coordinate boundary.
Coordinate-dependent operations assume an admitted AST identity, frame, topology, detector binding, support, validity, and uncertainty where required. The absence of such authority does not license inference or clamping.

SCI-RTC-ASM-004 – Calibration boundary.
Calibration-factor production, target-atmosphere correction, beam meaning, uncertainty, and promotion belong to SCI-BEAM/SCI-CAL. RTC remains in raw $\Delta f/f$ through replacement, filtering, and sampling. Atmospheric templates are diagnostic evidence for leakage, Tune/readout health, shift and artifact assessment, and selection/review only; v0.1 does not subtract them from science x. SCI-CAL consumes only the exact RTC x product and applies absolute `flxscale` plus target atmosphere only after RTC.

SCI-RTC-ASM-005 – Selected-policy boundary.
Despike, level-shift detection, transition masks, segmentation, reset, plateau use, donor, mask, FIR, IIR, notch, atmospheric-template diagnostics, leakage, state, edge, non-finite, recovery, sampling, diagnostic, and failure choices are operation classes admitted across application contexts and governed by selected versioned policies. Admission does not mean execution; a class runs only when the resolved plan selects and numerically resolves it. No current default or numerical threshold is promoted to a universal scientific constant by this contract.

SCI-RTC-ASM-006 – Conditional linearity.
Linear response and covariance identities condition on fixed paired-stream selection and realized state. At fixed state the conditioned-x numerical Jacobian has a zero r branch. Cross-coordinate covariance and shared selectors are retained in selection, learned-parameter, model, and cross-term uncertainty. Data-derived selectors make the unconditional operation generally nonlinear and nonstationary, and a total derivative may be unavailable at a selector boundary.

SCI-RTC-ASM-007 – Statistical input.
Moments or covariance are claimed only from a supplied statistical model with declared included and excluded components, correlations, and limitations. Absence does not imply diagonal, white, stationary, independent, or zero covariance, and a qualified partial claim need not pretend to be total.

SCI-RTC-ASM-008 – Learned-mode scope.
Every learned-then-fixed RTC operation uses one declared learning population, one resolved immutable plan, and one explicit compatibility domain. The first learned sampling scope produces one common observation plan. Per-array, per-scan, heterogeneous-cadence, noise-aware, and continuously adaptive sampling execution are successor scopes. Bounded iterative notch-plan refinement is admitted only as a finite outer sequence of actual attempts, bounded by a separately configured maximum, over complete immutable accepted plans; rejected attempts do not advance the accepted-plan sequence, unused maximum attempts are not artificial no-ops, and no parameter evolves within an apply pass.
SCI-RTC-ASM-009 – Analytical authority.
Exact analytical source-times-realized-filter transfer and phase-zero alias accounting define learned-plan admission. Deterministic vectors may falsify a realization, while source injections and observational studies are required for separately claimed performance or science-impact evidence.
SCI-RTC-ASM-010 – Compact representation.
Dense observation-sized response, covariance, or per-sample provenance is not required when a factorized, sparse, interval-based, or generative representation exactly reconstructs the operator, support, causes, and exceptions. Normal despike output may retain compact treatment state and population counts and characteristics without a complete per-event or per-donor manifest; detailed event fits and donor links may be a separately requested verbose or diagnostic product whose presence is numerically inert.
SCI-RTC-ASM-011 – Downstream authority.
RTC supplies causal inputs but does not own PTC cleaning/weights, VAL eligibility policy, MAP estimation, BEAM inference, empirical NOI products, MAP-003 response tracing, map filtering, source fitting, or fruit-loop recurrence.
SCI-RTC-ASM-012 – Claim separation.
The contract is an algebraic and semantic authority only. It contains no claim that a present implementation conforms, that a representation is faithful, that observational performance is adequate, or that a mode is ready for production.

4.1 Explicit non-goals

This revision does not derive ALIGN timing, AST coordinates, SCI-BEAM or SCI-CAL calibration, PTC, VAL, MAP, or FLT estimators, source fitting, or recurrence; authorize absolute-timing or astrometric corrections; redesign mature despike/FIR/IIR/notch numerics; choose universal numerical thresholds; authorize online adaptation; activate polarimetry; authorize atmospheric or common-mode template subtraction from science x; authorize automatic r-to-x correction, r calibration, use of r as an x donor, or a gain-, responsivity-, or general response-changing level-shift model; prescribe source classes, files, storage schemas, or product names; authorize per-array/per-scan cadence; inspect implementation; perform validation or reductions; or assert production status.

5 Sequential normative requirements

The following identifiers are stable within v0.1/r0.9. Conformance is requirement-by-requirement; satisfying a neighboring requirement does not waive another one.

Area	ID	Normative statement
Boundary	`SCI-RTC-REQ-001`	RTC shall admit only exact aligned `xs`/`rs` pairs whose signals are raw, uncalibrated detector coordinates and whose ordered pair identity, assigned grid, support, origin/synthesis state, and independently declared validity are present; a missing partner is malformed required input.
Boundary	`SCI-RTC-REQ-002`	Every requested RTC application context shall declare requested use, paired input and interval, x/r sign and reference/baseline, valid domain, named consumers, and constraints while retaining raw coordinate identity; a context label shall neither restrict admitted operation classes nor give either member a calibrated or Stokes meaning.
Boundary	`SCI-RTC-REQ-003`	Every RTC application context shall produce conditioned raw x as its required numerical output and retain the exact paired raw r parent plus every causal r diagnostic, selector, segmentation, and resolved-plan use in the consumer-neutral atomic bundle; a conditioned r output is optional unavailable unless a separate channel-specific operator is authorized.
Boundary	`SCI-RTC-REQ-004`	A calibrated detector signal shall be a distinct downstream SCI-CAL product in the top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention, with admitted factor, atmosphere, detector, validity, uncertainty, and provenance state.
Boundary	`SCI-RTC-REQ-005`	RTC shall preserve the exact paired raw parent, conditioned-x plan/response/support, factor authority, atmospheric diagnostic lineage, and distinct SCI-CAL atmosphere lineage required downstream; atmospheric templates may inform diagnostics, selectors, or review but shall not be numerically subtracted from science x in v0.1.
Identity	`SCI-RTC-REQ-006`	Observation, Tune, array, network/interface, detector occurrence, APT binding, native row, aligned slot, scan, science interval, context interval, output slot, stage, and product identity shall remain distinct and immutable.
Identity	`SCI-RTC-REQ-007`	The ordered detector and sample shapes, index bases, native-to-aligned mapping, aligned-to-output map, scan membership, and memory order shall be explicit; indices shall not serve as cross-artifact detector identity.
Identity	`SCI-RTC-REQ-008`	RTC shall preserve the admitted ALIGN cadence, phase, rate, gaps, support, and synthesis/origin state without claiming physical integration-event timing or absolute epoch authority.
Identity	`SCI-RTC-REQ-009`	A coordinate-dependent operation shall consume explicit AST identity, frame, topology, detector binding, support, and validity required by its resolved plan, with no inference, clamping, or implicit frame conversion.
State	`SCI-RTC-REQ-010`	RTC application context, immutable resolved plan, and realized record shall be distinct one-way objects; requested, effective, observation-resolved, learned, resolved, and realized substates may refine them, but no later object or substate shall overwrite an earlier one.
State	`SCI-RTC-REQ-011`	Outer, inner, full, mini, diagnostic, simulated, processed, and any other produced stage roles shall have distinct immutable identities and explicit parent/processing links without requiring duplicate computation.
Policy	`SCI-RTC-REQ-012`	Despike, spike-evidence, level-shift learning, segmentation, plateau-assessment, leakage-diagnostic, donor, replacement, mask, filter, state, edge, non-finite, recovery, sampling, flag, diagnostic, and fallback classes are admitted across RTC application contexts. Availability shall not be confused with enabled, selected, numerically resolved, or scientifically qualified; every executed action shall be governed by an admitted versioned resolved plan rather than an undocumented default.
Order	`SCI-RTC-REQ-013`	RTC shall begin at the admitted upstream-aligned paired raw parent, perform spike-candidate masking/downweighting for level-shift learning on that original pair, resolve finite transition support and stable plateaus, apply any selected and valid additive x plateau correction, then apply compatible x donor replacement within stable segments, temporal filtering, and phase-zero sampling before SCI-CAL consumes only x and applies absolute `flxscale` and target-atmosphere correction as in equation (SCI-RTC-EQ-004); no calibrated RTC branch or second ALIGN application is authorized.

Area	ID	Normative statement
Donor	`SCI-RTC-REQ-014`	A raw donor q to raw target d using the detector-static comparison coordinate $\chi_i = \phi_i x_i$ shall use ϕ_q/ϕ_d only when both `flxscale` factors are valid for the exact occurrences under the same compatible convention/domain and $\phi_d \neq 0$.
Donor	`SCI-RTC-REQ-015`	The raw donor route in REQ-014 shall use `flxscale` and shall neither require nor restore a scientific role for legacy `responsivity`.
Donor	`SCI-RTC-REQ-016`	When REQ-014 is not available, replacement shall fail under that route unless a separate donor-to-target authority declares direction, quantity/unit domain, support, uncertainty, validity, and provenance.
Donor	`SCI-RTC-REQ-017`	When despiking is selected, RTC shall actually apply the selected replacement or recovery to each accepted target x cell, or record the selected explicit failure/no-correction disposition. The resolved detector, target, eligibility, topology, score/tie, donor-weight/time, source-mask/flag, and fallback policies shall be reconstructible. Normal production output shall retain compact treatment state and useful spike-population counts and characteristics; complete event-by-event fits and realized donor links are required only for a separately selected verbose or diagnostic product.
Donor	`SCI-RTC-REQ-018`	A replaced target shall retain target identity, link every donor and factor, preserve the replacement cause, and expose correlations from reused or shared donors; finiteness shall not create independent exposure.
Influence	`SCI-RTC-REQ-019`	An output whose exact representative source occurrence $\rho_{dn} = (d, Mn)$ is ALIGN-synthesized shall be universally excluded as an independent detector measurement; nonrepresentative synthesized influence shall preserve its numerical effect, typed cause, and support for the downstream consumer's declared eligibility policy.
Influence	`SCI-RTC-REQ-020`	An output whose exact representative source occurrence $\rho_{dn} = (d, Mn)$ is RTC-replaced shall be universally excluded as an independent detector measurement; nonrepresentative replacement influence shall preserve its numerical effect, typed replacement cause, declared donor-influence class, and support for the downstream consumer's declared eligibility policy.
Filter	`SCI-RTC-REQ-021`	Every enabled response-changing stage shall record chronological order, exact full-precision coefficients or exact digest, coefficient convention, normalization, numeric precision, rate, direction, state, context, mask, edge, non-finite policy, exceptions, and the coherent segment bounded by gaps, resets, or level shifts.
Filter	`SCI-RTC-REQ-022`	An FIR claim shall expose exact impulse support, DC gain, phase convention, and normalization; a label such as "normalized" shall not substitute for the coefficient sum or response.
Filter	`SCI-RTC-REQ-023`	An IIR or notch claim shall expose an exact recurrence or state-space equivalent, realized poles/zeros where applicable, initial/final state, reset boundary including admitted level shifts, transient/guard rule, precision, and stability status; reset is ordinary at an accepted shift, and carry is permitted only as separately authorized continuity with explicit response and residual-contamination criteria.
Mask	`SCI-RTC-REQ-024`	A processing mask shall be represented as an operator control, including geometry or selection, boundary convention, dilation, denominator/segment behavior, and state interaction; it shall not be treated as acquisition validity or confidence.
Mask	`SCI-RTC-REQ-025`	Invalid, unavailable, wrong-frame, out-of-domain, or ambiguously bound coordinates shall not become outside-source samples and shall fail the affected required coordinate operation.
Finite	`SCI-RTC-REQ-026`	A non-finite required sample, factor, coefficient, coordinate, or filter state shall be rejected with a typed cause, marked unavailable with that cause over its exact transitive influence, or handled by an explicitly authorized prior replacement/recovery rule; it shall not be silently coerced to zero or encoded only as an undifferentiated generic non-finite.
Edge	`SCI-RTC-REQ-027`	Boundary extension, incomplete context, filter guards, IIR recovery, short/empty scans, and output selection shall be explicit; an edge or incomplete-state cell shall not be labeled an ordinary interior sample.

Area	ID	Normative statement
Sampling	SCI-RTC-REQ-028	V0.1 downsampling shall use only phase-zero point selection $y^{x,\mathrm{RTC}}[n] = v^{x,\mathrm{preD}}[Mn]$ from the final pre-decimation x stream; the representative source occurrence is exactly (d, Mn) on the raw aligned paired parent before replacement/filtering, and arithmetic-mean or other block-aggregate downsampling is excluded.
Sampling	SCI-RTC-REQ-029	Each coherent segment shall record M, zero phase, input/output rates and grids, the cardinality of equation (SCI-RTC-EQ-011), selected-point representative time, full transitive support, edge disposition, and flag/influence aggregation.
Sampling	SCI-RTC-REQ-030	Any alias-free or bounded-alias claim shall include the exact realized prefilter and all folded bands of equation (SCI-RTC-EQ-014), with a declared input band, metric, bound, and unavailable state; point selection alone shall not be called anti-alias filtering.
Sampling	SCI-RTC-REQ-031	Fixed mode shall resolve an admitted factor/filter/rate/compatibility plan before application and record it independently of realized output.
Learned	SCI-RTC-REQ-032	Every learned RTC operation shall preserve distinct requested, effective, observation-resolved, learned-evidence, resolved-plan, and applied states and shall not advertise a plan before admitted evidence and numeric policy are complete.
Learned	SCI-RTC-REQ-033	Learned resolution shall choose the largest admitted integer factor only after astronomical-transfer, exact phase-zero alias, sampling, filter-realizability, and downstream-compatibility constraints all pass using the smallest admitted beam and maximum valid in-scan speed.
Learned	SCI-RTC-REQ-034	The first learned applied scope shall use one common immutable observation cadence and filter across arrays and scans; percentile speed shall remain diagnostic rather than safety authority.
Learned	SCI-RTC-REQ-035	Apply shall consume exactly one immutable resolved plan and shall not retune, adapt, or silently fall back; an unavailable or rejected plan shall follow the selected failure/fallback policy before apply begins.
Learned	SCI-RTC-REQ-036	Restart shall restore a state-complete resolved plan, verify exact plan/input compatibility, and reject stale or incompatible beam, speed, scan, coefficient, rate, phase, application context, consumer binding, or parent identity.
Response	SCI-RTC-REQ-037	The atomic bundle shall contain the complete RTC-local realized response from the admitted aligned x parent, or a typed unavailable state; optional end-to-end response to native acquisition coordinates shall be separately identified and explicitly composed with admitted upstream ALIGN response. A partial filter kernel or silently repeated ALIGN operator shall not be labeled the conditioned response.
Response	SCI-RTC-REQ-038	Selection, detector mixing, masking, finite boundaries, state, or multirate behavior shall use a local/factorized detector-time response or truthful unavailability rather than an unjustified scalar transfer; fixed-state $x \leftarrow r$ numerical response is zero, cross-coordinate selection dependence is separate, and downstream SCI-CAL response remains a separate composition.
Response	SCI-RTC-REQ-039	A scalar LTI response shall be claimed only on a stated uniform, fixed-state, detector-separable, translation-invariant interior domain with no data-derived selector or undeclared rate change.
Response	SCI-RTC-REQ-040	Where defined, response shall distinguish amplitude, power, phase, group delay, sampling rate, Fourier convention, alias images, support, and response uncertainty; undefined phase/delay at a response zero shall remain unavailable.
Response	SCI-RTC-REQ-041	RTC-local output support shall be expanded transitively from the admitted aligned grid through replacement, every filter, state boundary, edge rule, and phase-zero selection; optional end-to-end support shall additionally compose admitted upstream ALIGN support. A selected center index shall not stand in for full support.

Area	ID	Normative statement
Statistics	SCI-RTC-REQ-042	At fixed realized state, conditional numerical mean and covariance shall follow equations (SCI-RTC-EQ-016a)–(SCI-RTC-EQ-016b) with $J_{\mathrm{num},\Omega} = [L^x_\Omega\ 0]$, so only Σ_{xx} enters direct propagation. Every resulting claim shall name this conditioning and explicitly state which cross-coordinate, donor-reuse, shared-selector, overlap, learned-parameter, model, diagnostic, and cross-term effects are included and excluded.
Statistics	SCI-RTC-REQ-043	When selection depends on modeled random data, each uncertainty claim shall state whether selection bias, selection covariance, and correlations with retained terms are included, excluded, or unavailable; they shall not be set to zero merely by conditioning on the realized selector.
Statistics	SCI-RTC-REQ-044	Every RTC covariance or uncertainty claim shall explicitly name the components and correlations it includes and excludes, their support and basis, and the availability of calibration, atmosphere, beam, donor-scale, coefficient, cadence/phase, response, selection, and model nuisance terms and applicable cross terms.
Statistics	SCI-RTC-REQ-045	Unknown covariance components or correlations shall remain unavailable and shall not be replaced by diagonal, white, stationary, independent, zero, or scalar-weight assumptions. Truthfully labeled conditional or component-limited claims are permitted; only declared complete coverage may be called full precision or total uncertainty.
Validity	SCI-RTC-REQ-046	Acquisition support, independent x validity, independent r validity, pair identity, ALIGN synthesis, numerical computability, coordinate validity, flags/causes, operator masks, replacement, transitive influence, complete support, response status, consumer eligibility, uncertainty availability, and provenance shall remain separate states.
Validity	SCI-RTC-REQ-047	A flag shall describe a cause and shall not automatically become a mask, invalidity, weight, or universal eligibility decision; aggregation shall preserve every contributing cause required by the selected policy.
Output	SCI-RTC-REQ-048	RTC shall produce the consumer-neutral atomic bundle of equation (SCI-RTC-EQ-022): conditioned x, the immutable paired raw r parent, and every required common diagnostic/selector/lineage field. Consumers bind the members and optional diagnostic detail they need; neither conditioned x TOD alone nor an independently emitted r diagnostic is the complete required product.
Output	SCI-RTC-REQ-049	Required malformed identity/state or missing authority shall propagate to atomic required-output failure; optional unavailable provenance detail shall be typed and numerically inert.
Output	SCI-RTC-REQ-050	Compact provenance shall reconstruct the exact native IQ-to-x/r mapping, upstream ALIGN relation, admitted aligned pair identity, RTC-local input/parent identity, detector binding, grids and intervals, downstream CAL handoff authority, resolved donor policy and compact spike treatment/population summaries, level shifts/segments/resets, ordered filters, masks, coefficients/state/edges, sampling, local and optional end-to-end response/uncertainty status, exceptions, counts, and the one-way application-context/plan/record lifecycle. Detailed spike-event and donor-link manifests are separately requested diagnostic detail.
Output	SCI-RTC-REQ-051	Each diagnostic, including spike-population, leakage, and level-shift products, shall name estimator, unit, support, mask/selection, validity, availability, scalar-versus-frequency status where applicable, and whether it is inert, advisory, or enters the selected policy; toggling optional event-by-event or donor-level diagnostic detail shall not change values, flags, influence, eligibility, cardinality, response, or failure.
Consumer	SCI-RTC-REQ-052	Beyond the direct exclusions in REQ-019–020, RTC shall supply cause-preserving transitive influence rather than a universal downstream eligibility bit; each consumer shall bind the exact RTC parent, declare its influence-to-eligibility policy, and shall not erase causes or strengthen unavailable timing, coordinate, calibration, response, covariance, or eligibility claims.
Consumer	SCI-RTC-REQ-053	PTC-disabled terminal RTC output shall be a valid requested state when selected; no PTC or map product shall be implied, while any requested downstream product shall require the exact admitted RTC bundle.
Claims	SCI-RTC-REQ-054	Algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, and production readiness shall be stated and assessed as separate claims.

Area	ID	Normative statement
Lifecycle	`SCI-RTC-REQ-055`	Every learned operation shall name its learning population, estimator, support, selection, exclusions, quality, uncertainty, validity, and compatibility domain; same-observation learning shall be identified as data-dependent.
Lifecycle	`SCI-RTC-REQ-056`	Resolve shall convert evidence and fixed policy into one immutable plan containing every applicable sequence, family, coefficient identity, rate/convention, order/taps, band/notch parameter, applicability, mask, donor, edge/guard/reset, decimation/timing, response/support, uncertainty, validity, and downstream-compatibility field.
Lifecycle	`SCI-RTC-REQ-057`	A plan shall be resolved only when every required predicate and tolerance passes; missing evidence, missing tolerance, rejected evidence, or unavailable compatibility shall not be treated as pass.
Lifecycle	`SCI-RTC-REQ-058`	Any coefficient or operator parameter that evolves during apply shall be governed by a separately authorized online-adaptive estimator contract; it shall not be represented as learned-then-fixed filtering.
Design	`SCI-RTC-REQ-059`	Every despike or spike-evidence, level-shift-learning, segmentation, plateau-assessment, leakage-diagnostic, donor, mask, notch, low/high/band-pass, FIR, IIR, decimation, edge/state, or non-finite operation shall declare scientific purpose, signal/contaminant model, mathematical operator, design parameters, learn/resolve/apply behavior, astronomical and calibration consequences, validation/falsification, and unavailable states.
Design	`SCI-RTC-REQ-060`	Filter admission shall use the exact source-times-realized-response over the declared scan-speed/direction, projected-beam, source-class, prior-response, and cadence domain; $f \sim v/\theta$, a scalar FWHM, or a nominal cutoff shall remain pedagogical only.
Notch	`SCI-RTC-REQ-061`	Every notch shall resolve center, rate/frequency convention, width or Q, depth, family/order, poles/zeros where applicable, phase/group delay, state, transient/guard, applicability, learned uncertainty, drift tolerance, and combined response for overlaps.
Bands	`SCI-RTC-REQ-062`	Every admitted low-, high-, or band-pass shall resolve pass/transition/stop bands, ripple, attenuation, phase/delay, family, order/taps, precision, edges/guards, retained astronomical modes, and unavailable modes; no class label shall substitute for exact response.

Area	ID	Normative statement
FIR	`SCI-RTC-REQ-063`	FIR order or tap count shall follow owner-resolved transfer, alias, phase, support, guard, precision, and realizability constraints such as equation (SCI-RTC-EQ-025); an arbitrary integer or approximate family formula shall not be scientific authority.
Donor	`SCI-RTC-REQ-064`	Donor replacement shall be identified as a continuity surrogate at the target identity and shall separately establish numerical convention transfer, donor admissibility, and scientific eligibility; compatible `flxscale` shall not be evidence of common sky signal.
Donor	`SCI-RTC-REQ-065`	Donor scaling in a Beammap application context shall use an explicitly identified prior APT or other pre-existing calibration authority and shall not use a new `flxscale` derived from the same Beammap conditioning lineage.
Sampling	`SCI-RTC-REQ-066`	Decimation factor and prefilter shall be justified from cadence, exact response, folded-band alias bounds, smallest projected crossing time, maximum valid scan speed, source class, timing/coordinate accuracy, and downstream compatibility.
Timing	`SCI-RTC-REQ-067`	Each plan shall resolve temporal registration through zero phase, shifted output time, coordinate correction, response-aware downstream use, or an approved bound on residual centroid/PSF effect; selected grid time alone shall not imply response centering.
CAL	`SCI-RTC-REQ-068`	Calibration transfer between Beammap and target processing shall bind the complete RTC plans and require matching response or explicit calculation, renormalization, or observational evidence; identical operation labels or preserved units are insufficient.
Validation	`SCI-RTC-REQ-069`	Validation of a notch, broad-band/FIR, donor, or decimation plan shall vary its scientific design parameters over the admitted scan/beam/source domain and measure astronomical transfer, phase/centroid, PSF/morphology, ringing, support/edge loss, covariance, and alias or interference residual as applicable.
Claims	`SCI-RTC-REQ-070`	No plan shall be called science-qualified solely because coefficient algebra, optimization, serialization, or deterministic operator fixtures pass.

Area	ID	Normative statement
Iteration	`SCI-RTC-REQ-071`	Iterative notch-plan refinement shall use a finite owner-approved maximum attempt bound and stopping policy; actual attempt count, configured maximum, and accepted-plan count shall remain distinct, unused maximum attempts shall not become artificial no-ops, and an unbounded repeat-until-satisfactory process is unavailable.
Iteration	`SCI-RTC-REQ-072`	Every apply of accepted plan Π_k shall consume exactly that immutable plan and shall not change coefficients, notch parameters, policy, masks, donor rules, decimation, or state conventions.
Plan	`SCI-RTC-REQ-073`	Every proposed successor $\widetilde{\Pi}_a$ shall be a complete cumulative plan that dispositions all prior and proposed notches and records all stages, order, coefficients/precision, state, masks, edges/guards, intervals, decimation/timing, response/support, uncertainty, validity, and compatibility; rejection shall not assign it an accepted-plan index.
Replay	`SCI-RTC-REQ-074`	The default evaluation and final production rule shall apply each complete accepted plan to original admitted $x^{(0)}$ as in equation (SCI-RTC-EQ-029); exploratory attempts shall not accumulate hidden rounding, padding, state, duplicate-filter, mask, flag, or parent effects.
Cascade	`SCI-RTC-REQ-075`	A cascade from a preceding filtered product shall require explicit scientific-operator authority and immutable intermediate parents, with total response, state, phase, edges, support, covariance, uncertainty, and sequential parentage retained.
Evaluate	`SCI-RTC-REQ-076`	Each successor attempt shall learn from the current accepted-plan evaluation and assess intended line attenuation and residual, parameter error/drift, predicted-versus-measured response, stability, ringing/transient, edge/guard loss, phase/delay, source amplitude/centroid/PSF, extended transfer, covariance, and support where applicable.

Area	ID	Normative statement
Candidate	SCI-RTC-REQ-077	A successor line shall be admitted only after comparison of original, predicted, and conditioned spectra, cumulative response, line uncertainty, astronomical spectrum, transfer/alias budgets, and detector/network/array/scan/interval consistency excludes notch-edge, ringing, transient, leakage, normalization, numerical, and response-floor artifacts.
Budget	SCI-RTC-REQ-078	Successor admission shall preserve owner-approved cumulative interference improvement, removed bandwidth, astronomical transfer, source amplitude/centroid/PSF, phase/registration, ringing/transient, guard/support, stability, cadence/response compatibility, uncertainty, and provenance budgets.
Stability	SCI-RTC-REQ-079	Plan stability shall compare the complete line set and matching identity, centers, widths/Q, depths, family/order, precision, applicability, state/reset, masks/guards, broad-band stages, decimation/timing, response/support, and compatibility over the required consecutive accepted-plan comparisons.
Stop	SCI-RTC-REQ-080	Every attempt shall record one typed accepted, rejected, or stop disposition plus its cause; maximum-attempt, oscillating-plan, and nonconvergent-parameter terminations shall not be labeled convergence or plan stability, and rejection shall leave the last accepted plan available unless policy says otherwise.
Lineage	SCI-RTC-REQ-081	Every attempt shall preserve attempt index, current accepted-plan index, original input, previous accepted plan, learning parent/population and PSD definition, original/predicted/measured spectra, proposal and disposition, predicates/tolerances, any newly accepted complete plan/evaluation product, cumulative response/astronomical diagnostics, stop cause, and final plan identity in one-way state.
Restart	SCI-RTC-REQ-082	Serialization and restart shall reproduce the final selected complete plan and declared parent rule exactly and reject incompatible original input, rate, learning population, parent, coefficient, interval, state, or policy identity.

Area	ID	Normative statement
Pair	SCI-RTC-REQ-083	Every required admitted aligned occurrence shall contain both x^A_{dj} and r^A_{dj} with exact common detector, sample, stream, IQ-mapping revision, and ALIGN lineage; a missing member shall fail required-input admission rather than be synthesized, copied, or zero-filled.
Pair	SCI-RTC-REQ-084	Paired identity shall remain exact through selection, segmentation, conditioned-x filtering/sampling, bundle construction, serialization, and restart; retaining raw r parentage does not imply that the x numerical operator was also applied to r, and reordering or independently joining members is not equivalent.
Mapping	SCI-RTC-REQ-085	The upstream native $IQ \to xr$ mapping and subsequent ALIGN relation shall expose distinct authority, revision, transform identity, any local coefficient/offset/Jacobian convention, applicability, validity, support, and provenance for each admitted occurrence; RTC shall consume the resulting aligned pair rather than infer, linearize globally, repair, or reapply either upstream mapping.
Validity	SCI-RTC-REQ-086	Numerical and scientific validity of x and r shall be represented independently while retaining the pair; invalidity of one member shall not silently invalidate, replace, or validate the other.
Leakage	SCI-RTC-REQ-087	Optical response in r shall not be assumed zero. Any leakage claim shall identify source class, estimator, support, selection, validity, uncertainty, x/r units or scales, mapping revision, normalization, metric, Tune/loading/epoch domain, ratio dimensionality, and response convention.
Leakage	SCI-RTC-REQ-088	An atmosphere-derived x/r leakage diagnostic shall identify the atmospheric template or population, temporal/frequency support, masks, estimator, quality, uncertainty, applicability, shared-data dependence, noise-in-both-coordinates treatment, and detector self-inclusion rule independently of any bright-source result; correlation or direct noisy r-on-x regression alone is not an unbiased coefficient claim.
Leakage	SCI-RTC-REQ-089	A bright-source-derived x/r leakage diagnostic shall identify the source, model, spatial/temporal support, masks, estimator, quality, uncertainty, and applicability independently of any atmospheric result.
Leakage	SCI-RTC-REQ-090	Atmosphere-derived and bright-source-derived leakage products shall retain separate parentage and shall not be averaged, substituted, or promoted as mutually validating without an explicitly authorized comparison model.

Area	ID	Normative statement
Leakage	SCI-RTC-REQ-091	Each leakage product shall state whether it is scalar, band-limited, or frequency dependent and shall provide the corresponding response/support and coordinate-comparison compatibility, or a typed unavailable state. A ratio or angle shall not be compared across detectors, Tunes, loading states, epochs, or mapping revisions unless coordinate conventions are compatible or an explicit transformation and metric are applied.
Boundary	SCI-RTC-REQ-092	V0.1 shall not automatically correct x using r, calibrate r, use r as an x donor, or treat a valid r coordinate as a substitute for an invalid x coordinate.
Statistics	SCI-RTC-REQ-093	Any joint x/r selector, estimator, learned parameter, or diagnostic shall preserve cross-coordinate covariance, common support, shared selection, and unavailable components; independence shall require evidence. Such dependence shall not be misreported as a fixed-state $r \to x$ numerical branch, and a total derivative may be unavailable at selector boundaries.
Shift	SCI-RTC-REQ-094	Every accepted level-shift disposition shall retain compact event identity and treatment state, network timing reference, detector-specific finite transition interval or conservative bound in physical time, timing-vector-derived sample support, coherence tolerance, x and r evidence or unavailability, additive-offset status, quality/uncertainty status, detection state, and cause. Normal production output may summarize event populations; event-by-event fits and detailed diagnostics are required only when the selected verbose/diagnostic product requests them.
Shift	SCI-RTC-REQ-095	Level-shift learning shall use the original aligned paired parent while identified isolated-spike candidates are excluded, masked, or robustly downweighted without donor substitution. Shift boundaries shall be resolved before any donor replacement; later x replacement shall remain within a resolved stable segment and no donor relation shall cross an unresolved or accepted shift boundary.
Shift	SCI-RTC-REQ-096	A selected level shift shall determine or conservatively bound a finite transition interval in physical time and derive its affected sample support from the applicable timing vector. Those cells shall be explicitly flagged as level-shift-transition affected, excluded from both plateau estimators, assigned to neither plateau, and remain scientifically unmodeled after any offset correction. Any downstream propagated influence or guard shall remain distinguishable from the underlying physical transition support.
Shift	SCI-RTC-REQ-097	A selected level shift shall delimit coherent processing segments and reset filter or estimator state as the ordinary behavior. Carry across the boundary is permitted only as separately authorized continuity with an explicit response and residual-contamination criterion; donor links shall never cross it.
Plateau	SCI-RTC-REQ-098	Any plateau diagnostic or additive-offset estimate shall declare coordinate, segment, estimator, stable support on each side, transition exclusion, masks, uncertainty, quality, validity, reference plateau, and whether it is inert, advisory, or a selected policy input.

Area	ID	Normative statement
Shift	SCI-RTC-REQ-099	Multiple level shifts shall retain ordered distinct event identities, including overlapping guards and short intervening segments, and shall follow a selected conflict and minimum-segment policy.
Leakage	SCI-RTC-REQ-100	When coordinate-specific preprocessing or filtering can change an x/r leakage estimate, pre- and post-operation diagnostics shall be separately identified and their common comparison domain, support, and response shall be declared.
Plateau	SCI-RTC-REQ-101	A resolved RTC plan may estimate one additive difference between sufficiently stable pre/post x plateaus and translate one valid plateau to the declared reference plateau. The plan shall specify estimator, required support, quality/uncertainty criteria, reference plateau, application direction, and failure behavior. It shall not apply the offset to transition support or make that support valid, and it shall not introduce a gain, responsivity, or other response-change parameter.
State	SCI-RTC-REQ-102	Pair admission, mapping, spike-candidate evidence, level-shift learning, physical transition support, propagated influence, segmentation, additive-offset estimation/application, post-segmentation x replacement, reset, plateau evaluation, and leakage diagnostics shall expose requested, effective, resolved, realized, unavailable, and failed states with typed causes as applicable.
CAL	SCI-RTC-REQ-103	SCI-CAL shall receive only the exact conditioned x member plus the complete paired RTC bundle needed for traceability; the r member shall not receive SCI-CAL calibration or target-atmosphere correction.
Atm.	SCI-RTC-REQ-104	Atmospheric templates in RTC v0.1 shall be used only for leakage estimation, Tune/readout-health diagnostics, level-shift or artifact evidence, and declared selection/review inputs; they shall not be subtracted from science x. Any future numerical common-mode or atmospheric-removal estimator requires separate PTC or successor authority.
Context	SCI-RTC-REQ-105	Every resolved application-context plan shall disposition paired mapping, x/r validity, retained raw r parentage, leakage diagnostics, level shifts, segmentation/reset, and diagnostic atmosphere treatment. All v0.1 operation classes remain admitted across contexts until evidence or explicit policy disqualifies them; only operations explicitly selected and numerically resolved execute. Consumers bind the same consumer-neutral bundle, and subsequent CAL, PTC, VAL, MAP, and FLT use follows their own contracts.

Area	ID	Normative statement
Plateau	`SCI-RTC-REQ-106`	A plateau shall be retained for a requested application context only when it contains enough valid support for every required operation and estimator in that context; support shall be evaluated against masks, source/atmosphere content, filter context/state, PSD/notch learning, source geometry, cadence, and downstream needs as applicable, not sample count alone. If stable pre/post support is insufficient for a defensible additive offset, RTC shall not invent one; the event remains a segment boundary and the resolved plan shall retain or reject each usable plateau explicitly.
Leakage	`SCI-RTC-REQ-107`	Stable plateaus immediately before and after an accepted level shift shall have their optical-leakage or more general paired optical-response states compared on declared common support, or the comparison shall be explicitly unavailable when adequate optical excitation is absent; evidence of a material response change shall make additive correction unavailable and shall not trigger a fitted gain-change model in v0.1. Gain- or response-changing interpretation remains successor scope unless observational evidence reopens this assumption.
Output	`SCI-RTC-REQ-108`	Any separately requested conditioned r product shall require an explicit r-specific numerical operator, unit, response, support, validity, uncertainty, application context, and provenance; exact pairing, shared selection, or an x operator shall not implicitly authorize or define that product.

6 Falsifiable predictions and limiting cases

Each prediction is an algebraic or semantic falsification target. A future conformance activity may choose an exact numerical tolerance justified by the operation and representation before observing candidate output. No result is reported by this document.

ID	Construction and prediction	Falsifier
`SCI-RTC-PRED-001`	Identity chain: paired raw application context, $A = I$, no replacement or filter, $M = 1$, valid original pairs. Both values, raw coordinate identity, pair order, cardinality, response, flags, and support are unchanged.	Missing/reordered partner, hidden calibration, value/order/cardinality change, flag loss, nonidentity kernel, or new claim of timing/coordinate authority.
`SCI-RTC-PRED-002`	Context neutrality: feed the same paired `xs`/`rs` input and resolved numerical plan to contexts historically labeled Beammap, Pointing, OOF, Science, and diagnostic-only. The same consumer-neutral bundle schema results; labels do not change operation availability or signal meaning, while consumers select the members they require.	A label silently enables, disables, or changes an operation; changes bundle shape or raw meaning; conditions r with the x operator; labels RTC output mJy/beam; or interchanges CAL and RTC output.
`SCI-RTC-PRED-003`	Raw donor direction: choose $\phi_q = 2$, $\phi_d = 4$, and donor $x_q = 3$. Equation (SCI-RTC-EQ-002) predicts raw target replacement 1.5 and equal detector-static comparison coordinates $\chi_q = \chi_d^R = 6$.	Use of ϕ_d/ϕ_q, output other than 1.5, unequal comparison coordinate, calibration claimed from χ, or dependence on legacy `responsivity`.
`SCI-RTC-PRED-004`	Raw donor availability: independently make either factor invalid, convention/domain-incompatible, occurrence-mismatched, or set $\phi_d = 0$. The `flxscale` route is unavailable.	Finite replacement, stale factor reuse, row-number-only binding, zero insertion, or success without separate donor authority.
`SCI-RTC-PRED-005`	Order: trace a paired raw input through spike-candidate evidence, level-shift segmentation, within-segment x replacement, every temporal stage, and phase-zero sampling, then trace only conditioned x through distinct downstream SCI-CAL while retaining the raw r parent.	Donor replacement precedes resolved boundaries or crosses one, atmospheric diagnostic evidence is subtracted, absolute calibration or target atmosphere appears inside RTC, r enters CAL, or the CAL parent omits the complete RTC plan.
`SCI-RTC-PRED-006`	Interior target/donor impulses under fixed selection. Each output column equals equation (SCI-RTC-EQ-012); donor energy appears in the declared detector column, support, scale, and phase.	Missing or circularly wrapped energy, unexplained mixing, wrong donor identity/direction, support, normalization, or phase.
`SCI-RTC-PRED-007`	Constant input with sufficient context. Output equals the constant times realized DC gain after the declared transient; a DC-preserving chain is exactly constant.	Drift, divide-by-zero mask normalization, edge values advertised as interior, or a false DC-preservation claim.
`SCI-RTC-PRED-008`	Interior step with complete context. Response equals the cumulative impulse response and settles according to DC gain and stability.	Undeclared overshoot, failure to settle, hidden state leakage, or disagreement with the impulse-derived response.
`SCI-RTC-PRED-009`	Affine ramp through each FIR and admitted alignment mapping. Output slope and offset follow the zeroth and first moments of the exact kernel.	Unexplained clipping, wrong phase/slope/offset, or a ramp-preservation claim without the required moments.
`SCI-RTC-PRED-010`	Long bin-centered and off-bin sinusoids at DC, passband interior/edge, transition, stopband, and near output Nyquist. Amplitude and phase follow equations (SCI-RTC-EQ-013) and (SCI-RTC-EQ-014).	Gain/phase outside preregistered tolerance, unrecorded frequency convention, or an aliased component labeled passband signal.
`SCI-RTC-PRED-011`	Sinusoids at the realized notch center and symmetric offsets under the exact rate and state. Depth, width, phase, and transient follow realized zeros/poles.	Center/depth/width mismatch, unstable pole, rate mismatch, hidden state, or an edge transient reported ordinary-valid.
`SCI-RTC-PRED-012`	Coordinate mask with valid inside/outside values plus invalid, unavailable, wrong-frame, out-of-domain, and ambiguously bound coordinates. Invalid classes fail the affected operation and remain distinct from outside.	Invalid becoming outside/unmasked, entering geometry, clamping to a center/edge, or loss of the distinct cause.
`SCI-RTC-PRED-013`	Samples just inside, exactly on, and just outside each selected mask boundary and temporal dilation. Membership follows the recorded inclusive/exclusive and dilation policy identically across execution modes.	An undocumented boundary, frame mismatch, asymmetric dilation, or sequence-dependent membership.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-014	Impulses and constants at first/last support cells and immediately inside complete FIR context. Values, guard, support, and output admission follow the selected edge rule.	Undeclared padding, truncation, wrap, renormalization, guard, or output selection.
SCI-RTC-PRED-015	Zero, impulse, and constant inputs split and rejoined at scan/chunk/state boundaries for an IIR. Results follow the declared reset/carry and terminal-state policy.	Prior-observation leakage, hidden carry/reset, unbounded non-finite poisoning, or contradiction between split and declared state semantics.
SCI-RTC-PRED-016	Lengths $0, 1, L-1, L, L+1$ around every support/guard threshold, with stable scan identity.	Crash, underflow, stale output, silent scan deletion/renumbering, or ordinary-valid status without required context.
SCI-RTC-PRED-017	Inject NaN, $+\infty$, and $-\infty$ separately into a sample, factor, coordinate, coefficient, and IIR state. Failure or unavailable influence follows REQ-026 exactly.	Silent zero/coercion, finite-looking output without cause, stale state recovery, or an unbounded tail reported valid.
SCI-RTC-PRED-018	Place synthesized and replaced causes first at the directly selected occurrence and then at noncenter members of full support. Direct cases are universally excluded; noncenter causes propagate unchanged and consumer eligibility follows the declared policy.	A direct case called independent, a noncenter cause erased or made universally ineligible by RTC, or consumer disposition reported without its policy.
SCI-RTC-PRED-019	For $M = 1, 2, 3, 4, 5$ and $N = 0$ through several incomplete final strides, enumerate $j = Mn$. Cardinality and selected time follow equation (SCI-RTC-EQ-011); no block mean appears.	Wrong N_y, dropped/duplicated point, nonzero phase, averaged value, timestamp misidentification, or rate mismatch.
SCI-RTC-PRED-020	Two sinusoids separated by an integer multiple of f_{out}, before and after the realized prefilter. Folded amplitude is the sum in equation (SCI-RTC-EQ-014).	A folded term missing from response, inadequate stopband labeled alias-free, or incorrect image frequency/amplitude.
SCI-RTC-PRED-021	Reuse one donor for two targets and give neighboring outputs overlapping filter support. Any covariance claim states whether the resulting cross-target and temporal correlations are included, excluded, or unavailable; when included they follow equation (SCI-RTC-EQ-016b) and the admitted covariance.	Undisclosed omission, excluded or unknown correlations treated as zero, independent weights asserted without evidence, or reused donor described as additional exposure.
SCI-RTC-PRED-022	Enable a response-changing mask, donor mix, stateful filter, edge, and decimator while providing only a scalar FIR transfer. Complete response state is unavailable.	The partial scalar transfer labeled complete, or downstream use that assumes omitted stages are identity.
SCI-RTC-PRED-023	Learned candidates include at least two safe factors and one failing each astronomical-transfer, alias, sampling, realizability, and compatibility constraint. Equation (SCI-RTC-EQ-021) chooses the largest passing factor.	A smaller safe factor selected without rule, any failing factor applied, a percentile speed used for safety, or a missing numeric predicate treated as pass.
SCI-RTC-PRED-024	Serialize and restart a learned plan, then alter one of beam identity, speed support, scan set, coefficient precision, rate, phase, application context, consumer binding, or parent. Apply rejects the mismatch.	Silent relearning, retuning, native fallback, stale-plan application, or a realized record rewriting the application context or plan.
SCI-RTC-PRED-025	Execute two observations sequentially with distinct rates, constants, factors, masks, donors, states, and output parents.	Any coefficient, state, factor, donor, mask, rate, flag, or product link leaks from the first observation.
SCI-RTC-PRED-026	Toggle the complete spike-event/donor manifest and other optional inert diagnostics while holding the compact treatment/population summary, resolved plan, and realized operator fixed. Values, flags, influence, eligibility, cardinality, response, failures, and normal summary are identical.	Despiking ceases to modify accepted targets, any scientific or numerical result changes, or an observational/production claim appears from the detail toggle.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-027	Bind identical learned parameters first to a separate training population and then to the apply observation. The numerical fixed-state response can match, but the second plan records data dependence and parameter/selection uncertainty.	Same-observation learning represented as fixed a priori, or conditional linearity used to erase selection or parameter uncertainty.
SCI-RTC-PRED-028	Remove in turn one required learning-population identity, tolerance, uncertainty, or downstream-compatibility predicate before resolve. The dependent plan remains unresolved and apply does not begin.	Missing evidence treated as pass, a default inserted, or output represented as applied learned mode.
SCI-RTC-PRED-029	Serialize one resolved plan and attempt during apply to alter a notch center, cutoff, tap count, donor rule, or decimation factor. The mutation is rejected or belongs to a separately authorized adaptive estimator.	Hidden retuning, silent fallback, or a realized record that rewrites the resolved plan.
SCI-RTC-PRED-030	Sweep notch center error, line drift, width, depth, Q/order, initialization, direction, and overlap. Residual line and astronomical amplitude, phase, centroid, covariance, transient, and edge loss follow the exact combined response.	Only line disappearance measured, or any changed astronomical observable omitted from the plan's validity evidence.
SCI-RTC-PRED-031	Scan the same projected source at multiple speeds and directions through a fixed filter. Its temporal spectrum and retained amplitude/PSF change according to equation (SCI-RTC-EQ-023) and exact response.	One nominal cutoff or scalar beam treated as speed- and direction-independent proof of preservation.
SCI-RTC-PRED-032	Design FIR candidates over multiple tap counts and coefficient precisions. The selected $N_\star$ is the smallest candidate passing every registered transfer, alias, phase, support, guard, and realizability predicate.	Arbitrary configured taps, approximate order formula promoted to authority, or a candidate with any failed predicate selected.

ID	Construction and prediction	Falsifier
`SCI-RTC-PRED-033`	Inject compact and extended sources through admitted high- or band-pass candidates. Low-frequency rejection produces the predicted lost broad modes, compensated response, or negative lobes.	Unit preservation labeled flux preservation, or removed absolute/extended modes reported available without response restoration evidence.
`SCI-RTC-PRED-034`	Choose donor and target with compatible `flxscale` but deliberately different sky signal or beam coupling. Raw-domain convention transfer remains algebraically correct while the surrogate is not a missing-sky reconstruction.	Compatible factors claimed to prove physical donor equivalence or independent exposure.
`SCI-RTC-PRED-035`	Attempt Beammap donor scaling with a new `flxscale` derived from that same conditioned Beammap. The route is rejected until an identified prior factor authority is supplied.	Circular use of the new factor or absence of the prior authority in lineage.
`SCI-RTC-PRED-036`	Apply filters with known nonzero group delay to Pointing/Beammap scans in opposite directions. Centroid displacement follows equation (SCI-RTC-EQ-027) with opposite scan-direction sign unless time/coordinates are corrected.	Selected grid time treated as response center, wrong displacement sign, or unrecorded correction.
`SCI-RTC-PRED-037`	Process calibrator and target with plans that share a nominal label but differ in coefficients, rate, phase, state, edge, or decimation. Cross-plan calibration transfer is unavailable absent explicit response accounting or observational evidence.	Nominal label, preserved unit, or `flxscale` alone treated as RTC-plan compatibility.
`SCI-RTC-PRED-038`	Pass all deterministic coefficient and operator fixtures but withhold the required source/scan design studies. Algebraic correctness may be supported; science qualification remains unassessed.	Deterministic pass promoted to observational performance, science qualification, or production readiness.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-039	Inject one strong coherent line and no other valid line. Attempt 1 accepts and suppresses it; Attempt 2 finds no acceptable successor. The accepted-plan count remains one and Π_1 is final with a no-successor stop.	A manufactured Π_2 or evaluation product, unbounded iteration, unsupported second notch, rejected evidence lost, or no typed stop/final-plan identity.
SCI-RTC-PRED-040	Inject two unequal coherent lines so the weaker becomes identifiable only after the stronger is suppressed. A later attempt accepts one complete two-notch plan as the next Π_k and replays it once on $x^{(0)}$; output follows the predicted cumulative response.	Attempt and accepted-plan indices conflated, increment-only plan, final use of an accumulated exploratory product, missing first notch, duplicate filtering, or response disagreement.
SCI-RTC-PRED-041	Create a peak at a first-notch edge that agrees with predicted cumulative ringing or response artifact. The successor candidate is rejected and the first plan remains final.	Artifact admitted as a line, response floor ignored, or prior accepted plan invalidated solely by rejection.
SCI-RTC-PRED-042	Compare applying a one-notch plan twice with applying the intended complete successor plan once to $x^{(0)}$. Values, response, phase, edges, support, and covariance differ unless exact operator equivalence is proved.	Repeated filtering treated as replay, or equality claimed from nominal notch identity without complete operator proof.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-043	Propose a successor that fails one cumulative transfer, stability, support, or compatibility predicate. The attempt is rejected; its evidence and cause persist while the accepted-plan index and final product remain unchanged.	Rejected proposal assigned a new accepted-plan index or evaluation, failed plan applied, prior product rewritten, rejected evidence erased, or missing predicate treated as pass.
SCI-RTC-PRED-044	Alternate between proposing addition and removal of the same line or vary its parameters outside declared stability across attempts. Termination is typed oscillating or nonconvergent.	Alternation or parameter instability labeled convergence, stable plan, or successful maximum-safe refinement.
SCI-RTC-PRED-045	Supply a sequence that reaches the owner-approved maximum attempt count without satisfying plan stability. The result records maximum-attempt termination and the last accepted plan, not convergence.	Maximum count labeled stable/converged, continued iteration, attempt count confused with accepted-plan count, or disappearance of the last accepted plan.
SCI-RTC-PRED-046	Serialize and restart the final cumulative plan, then vary original parent or learning-population identity. Exact compatible restart reproduces the product; either mismatch is rejected.	Silent relearning, stale-parent application, altered plan, or acceptance of incompatible learning lineage.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-047	Supply complete paired occurrences, then remove only one r partner and separately only one x partner. Complete pairs are admitted; either incomplete occurrence fails required-input admission.	Partner synthesis, copying, zero fill, independent join, or successful admission of an incomplete pair.
SCI-RTC-PRED-048	Apply a known native nonlinear $IQ \to xr$ mapping, then one admitted ALIGN relation, and serialize/restart. The aligned pair and both upstream identities reproduce equations (SCI-RTC-EQ-030) and (SCI-RTC-EQ-004); the RTC-local operator begins at that aligned pair.	A nonlinear mapping is forced into one global affine form, either upstream identity changes, or ALIGN is omitted or applied twice inside RTC.
SCI-RTC-PRED-049	Make x invalid and r valid, then reverse them. The pair persists and the two validity states remain independent.	One member's state copied to the other, invalid member used as valid, or pair discarded without selected policy.
SCI-RTC-PRED-050	Inject an optical signal with known nonzero r/x response, then rescale the r coordinate under an explicit mapping revision. Each estimator recovers the correspondingly transformed ϵ; a reported ψ is invariant only after the declared normalization and metric transform.	r assumed zero, a ratio or angle remains numerically unchanged under incompatible rescaling, or units, mapping revision, metric, uncertainty, or support are missing.
SCI-RTC-PRED-051	Give atmosphere and bright source deliberately different leakage ratios and compare them first under compatible and then incompatible coordinate conventions. Distinct products remain available; the incompatible comparison is unavailable until an explicit transform is applied.	Averaging, substitution, universal promotion, or cross-revision comparison without a compatible normalization and metric.
SCI-RTC-PRED-052	Generate frequency-dependent leakage and fit both a scalar and admitted spectral estimator. Only the spectral product supports the frequency-dependent claim; the scalar is range-qualified.	Scalar advertised as broadband authority, missing band/support, or absent response status.
SCI-RTC-PRED-053	Attempt to correct x from r, calibrate r, use r as an x donor, or replace invalid x with valid r. Every route is unavailable in v0.1.	Any route executes or is implied by shared pair identity.
SCI-RTC-PRED-054	Simulate correlated x/r noise and an r-dependent selector. At fixed state, $J_{\mathrm{num}} = [L^x \; 0]$ and direct covariance is $L^x \Sigma_{xx} (L^x)^\mathsf{T}$; across selector realizations, the joint model retains cross blocks and selection uncertainty or declares them unavailable.	A fixed-state numerical r branch appears, cross blocks are inserted directly into $L^x \Sigma_{xx} (L^x)^\mathsf{T}$, or selector dependence is discarded.
SCI-RTC-PRED-055	Add one network event with a known finite physical-time transition, bounded detector timing offsets, unequal x/r evidence, and adjacent isolated spikes; repeat it at different sampling cadences. Learning uses the original pair, reports one event identity, cadence-consistent timing-vector-derived transition support, stable plateaus, and only then permits valid additive correction and within-segment replacement.	A fixed sample count changes the inferred physical width, detector events lose their network parent, donor-repaired values drive the change point, shift is absorbed as spikes, amplitude is copied between coordinates, or a donor crosses the boundary.
SCI-RTC-PRED-056	Place isolated spikes immediately before and after a level shift and vary donor values independently. The resolved change point is invariant to unused donor values; replacement begins only after segmentation and stays inside each stable segment.	Change-point learning consumes donor substitutions, depends on donor policy, crosses a shift boundary, or circularly repairs evidence used to resolve that boundary.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-057	Put a filter impulse on each side of a declared level-shift boundary. The ordinary plan resets. A carry plan is admitted only with separately authorized continuity, complete response, and a passing residual-contamination criterion.	Silent cross-boundary state, ordinary carry without separate authority, failed/missing residual criterion, ordinary-interior labels in guards, or missing reset provenance.
SCI-RTC-PRED-058	Insert two level shifts with overlapping guards and a short middle segment. Event identities remain ordered and the selected conflict/minimum-segment policy gives one deterministic disposition.	Event collapse, order dependence outside policy, negative/ambiguous segment, or silent guard truncation.
SCI-RTC-PRED-059	Give stable pre/post plateaus a known additive offset around a finite transition. With correction disabled the values are unchanged; with a valid selected correction one plateau moves to the declared reference while transition cells remain flagged, unmodeled, and excluded.	An unselected correction changes values, the reference/sign is ambiguous, transition cells enter either plateau fit or become valid, the cause is erased, or a gain term is fitted.
SCI-RTC-PRED-060	Measure leakage before and after a filter whose x and r responses differ. The two diagnostics differ according to the composed response and retain separate identities.	One value reused across stages, filtering treated as commuting without proof, or pre/post parentage lost.
SCI-RTC-PRED-061	Hold the resolved x operator fixed and toggle an atmospheric leakage template between available and diagnostic-unavailable states. Conditioned x is unchanged while diagnostic availability and lineage change exactly as declared.	The template is numerically subtracted, changes x, silently becomes PTC cleaning, or is conflated with target SCI-CAL atmosphere.
SCI-RTC-PRED-062	Stop iterative refinement before $A_{\max}$ after A actual attempts. The record satisfies $K \leq A < A_{\max}$ and contains no fabricated attempts.	$A = A_{\max}$ forced, artificial no-op attempts, or accepted-plan count confused with either attempt quantity.
SCI-RTC-PRED-063	For Beammap, Pointing, OOF, Science, and diagnostic-only application contexts, inspect resolved plans and output bundles. The consumer-neutral schema retains conditioned x, raw r parentage, paired mapping/validity, leakage, level-shift/segment/reset, diagnostic atmosphere treatment, and consumer routing; optional conditioned r remains unavailable unless separately selected.	Missing disposition, context-dependent loss or shape change, the x operator silently applied to r, an operation enabled by label implication, or consumer routing that bypasses downstream contracts.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-064	Apply an ideal admitted mapping to a purely optical excitation whose nominal orthogonal projection is zero. The recovered r optical coefficient is consistent with zero within the preregistered estimator tolerance.	A nonzero coefficient outside tolerance, missing uncertainty, or a zero claim made without adequate excitation/support.
SCI-RTC-PRED-065	Apply a known small mapping rotation to the same excitation. The measured r/x leakage follows the predicted sign and magnitude over the declared local domain.	Wrong sign/magnitude, global-affine extrapolation outside the mapping domain, or rotation absorbed into an unrelated nuisance.
SCI-RTC-PRED-066	Generate shared optical signal with noise in both x and r, then compare direct r-on-x regression with the admitted estimator. Direct regression exhibits the predicted errors-in-variables or shared-data bias unless its method explicitly accounts for it.	Raw correlation promoted to an unbiased coefficient, self-inclusion ignored, or estimator uncertainty excludes shared-data dependence.
SCI-RTC-PRED-067	Scan an optical source whose expected r leakage follows the admitted source response across a bounded interval. The crossing remains a source/leakage event rather than an abrupt persistent level shift.	Expected optical leakage alone triggers a shift, creates a false plateau boundary, or loses source identity.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-068	Inject a gradual drift over declared support. It remains drift/nonstationarity evidence unless the selected change-point model independently supports an abrupt transition.	The drift is automatically collapsed into one step or an abrupt-shift label is assigned without model evidence.
SCI-RTC-PRED-069	Learn a PSD/notch plan first from data containing an unsegmented step and then from the same original data with the accepted transition and guards excluded. The first plan shows the predicted step contamination; the second excludes it under the declared support policy.	Identical stationary-learning claims without accounting, a step learned as a spectral line, or segmentation that still admits cross-boundary support.
SCI-RTC-PRED-070	Give stable pre/post plateaus materially different admitted leakage or paired optical-response states. The event remains distinct, additive correction is unavailable, and v0.1 fits no gain or responsivity change.	Additive equivalence is claimed despite the response change, a gain-changing correction is silently fitted, or pre/post comparison is omitted when support exists.
SCI-RTC-PRED-071	Construct one short plateau that supports readout-health diagnostics but not a defensible offset or stronger requested operations, alongside another usable plateau. The event remains a boundary, no offset is invented, and the plan separately retains or rejects each plateau for the application context.	One sample-count threshold controls all uses, an offset is fabricated, the entire scan is automatically discarded, or diagnostic validity is promoted to a stronger use.

6.1 Analytic limits

The following limits are required: a compatible donor pair with $\phi_q = \phi_d$ gives unit raw donor scale; a replacement indicator tending to zero gives identity; filter coefficients tending to a Kronecker delta give identity; $M \to 1$ gives no rate change; a source-mask radius tending to zero selects only exact valid coincidences under the declared boundary; complete context approaches the declared interior response; and a causal pole approaching the unit circle exposes increasing memory and never retains a stability claim at or outside the circle.

End of shared normative core: `SCI-RTC-v0.1/r0.9`. View-specific material cannot add to or modify this scientific authority.

A Informative conformance-record schema

This appendix supplies bookkeeping guidance only. It does not add a requirement. A review record should contain the following fields.

Field	Evidence content
Record identity	Stable review ID, date, reviewer role, exact contract revision, and supersession status.
Claim layer	Exactly one primary layer: implementation conformity, representation fidelity, observational performance, science impact, or production readiness.
Target identity	Exact source/build/artifact identity and any configuration or runtime identity needed to reproduce the observation.
Authority binding	One or more SCI-RTC requirement or prediction IDs; an equation/definition may be cited only through the shared core.
Preconditions	RTC application context, signal domain, detector occurrences, grid, intervals, immutable resolved plan, realized-record scope, and owner decisions required by the check.
Method	Static trace, deterministic vector, bounded matrix calculation, serialization inspection, failure injection, or other named method.
Expected result	Exact invariant and a tolerance justified before observing candidate output.
Observed result	Values, states, messages, artifacts, and exceptions without interpretation hidden in a generic pass flag.
Disposition	Pass, fail, not exercised, blocked by open owner decision, or evidence unavailable.
Claim limit	Explicit statement of what the record does not establish.

No status may be inferred from an omitted record. "Not exercised" means the evidence was not attempted; "blocked" means a named unresolved authority prevents a meaningful check; and "unavailable" means the attempted evidence cannot support the claim. These dispositions are not passes.

B Informative requirement-review routing

The table suggests evidence classes without prescribing implementation structure. Reviewers should use the exact crosswalk shipped with the package for one-to-one coverage.

Requirement range	Primary review surface	Useful evidence classes
REQ-001–005	Paired raw boundary and CAL handoff	Exact paired input, conditioned-x/raw-r-parent bundle inspection, x-only downstream CAL-parent failure injection, and atmospheric-diagnostic no-subtraction trace.
REQ-006–012	Identity, context–plan–record state, and selected policy	Serialization round trip, distinct-identity fixtures, one-way context/plan/record trace, context-label neutrality, unsupported-policy rejection.
REQ-013–018	Ordering, despiking, and donor transfer	Raw-RTC-then-CAL order trace, accepted-target modification check, compact spike-population summary, forced donor vectors, invalid-factor matrix, donor reuse/tie/fallback fixtures, and optional detailed-manifest toggle.
REQ-019–020	Direct exclusion and influence	Direct/noncenter synthesis/replacement injection, full-support cause propagation, downstream policy handoff inspection.
REQ-021–027	Filters, masks, edges, and finite policy	Coefficient/state serialization, impulse/step/constant/ramp/notch vectors, mask-boundary and non-finite failure injection.
REQ-028–031	Fixed phase-zero sampling	Enumerated length/factor cases, selected index/time identity, rate trace, exact folded-band calculation.
REQ-032–036	Learned plan lifecycle	Candidate decision table, maximum-safe selection, common-plan identity, immutable apply and restart mismatch fixtures.
REQ-037–041	Complete response and support	RTC-local factorization from the aligned parent, optional explicit ALIGN composition, local Jacobian fixtures, LTI-domain proof, edge/state/support expansion.
REQ-042–045	Statistical claim disclosure	Fixed-state $[L^x\ 0]$ covariance fixture; for every conditional, component-limited, or total claim, inspect explicit included/excluded components and correlations plus unavailable-state preservation.
REQ-046–051	Validity and consumer-neutral atomic output	Orthogonal-state fixtures, common bundle-schema inspection across contexts, atomic write/failure injection, compact provenance reconstruction, inert detailed spike/donor-manifest toggle.
REQ-052–054	Consumers and claims	Parent-binding inspection, unavailable-state preservation, disabled-PTC terminal path, claim-label audit.
REQ-055–058	Learn–resolve–apply lifecycle	Learning-population identity, predicate-completeness matrix, immutable-plan trace, mutation rejection, adaptive-estimator separation.
REQ-059–063	Scientific filter design	Purpose/model register, scan–beam transfer studies, notch and band specifications, constrained tap/order selection.
REQ-064–068	Donor, decimation, timing, calibration	Donor-meaning separation, circular-factor rejection, folded-band calculation, centroid/coordinate tests, cross-plan response comparison.
REQ-069–070	Validation and claims	Preregistered design sweeps and strict claim-layer disposition.
REQ-071–076	Refinement attempts and replay	Distinct attempt/accepted-plan trace, immutable apply, complete-plan reconstruction, original-input replay, explicit-cascade comparison, intended-consequence assessment.
REQ-077–082	Candidate, stability, stopping, provenance	Artifact discrimination, cumulative budgets, complete-plan stability, actual/max-attempt distinction, typed disposition, lineage, restart mismatch fixtures.
REQ-083–086	Pair admission, identity, mapping, validity	Missing-partner failure, mapping reconstruction, reorder/rejoin rejection, independent x/r validity matrix, restart identity.
REQ-087–093	Leakage, prohibited correction, joint statistics	Atmosphere/source estimator fixtures, coordinate rescaling and comparison compatibility, scalar/frequency status, forbidden-route rejection, fixed-state zero r branch, and joint selector covariance.
REQ-094–102	Level shifts, plateaus, and state	Original-pair spike masking/downweighting; network/detector finite physical-time transition support; timing-vector-derived sample masks; additive-only plateau model and optional valid offset correction; transition exclusion; post-segmentation donor fixtures; reset-by-default boundaries; separately authorized carry; compact production state; and typed lifecycle states.
REQ-103–105	CAL, atmosphere diagnostics, context and consumer routing	x-only CAL handoff, atmospheric-template no-subtraction checks, retained raw-r parentage, consumer-neutral bundle binding, all-class admission, explicit-selection execution, and context-label neutrality checks.
REQ-106–108	Plateau support, pre/post leakage, optional conditioned r	Application-context support matrix, insufficient-offset-support behavior, pre/post response comparison/unavailability, gain-model exclusion, and separate-r-operator admission failure.

C Informative response and representation checks

Review the response at the same granularity as the realized operation. A stored coefficient vector can support a component-level check; it does not by itself establish complete response. For each coherent segment, trace the ordered factorization, exact coefficients and precision, mask/state/edge exceptions, detector mixing, sampling phase, and support expansion to the artifact presented to consumers. Mark complete response unavailable if any enabled response-changing stage cannot be represented truthfully.

Begin that trace with exact general IQ-to-x/r mapping authority, one upstream ALIGN relation, and the admitted aligned RTC boundary. Reject a second ALIGN application in the RTC-local operator; compose it only for an explicitly identified end-to-end response. Exercise the conditioned-x operator while retaining raw-r parentage, zero fixed-state r numerical branch, and full cross-coordinate selector and uncertainty dependence. Verify ordinary reset at level shifts and admit carry only under separate continuity authority with response and residual criteria. Confirm that atmospheric templates change only diagnostic or declared selector state, not numerical science x. Reject any conditioned r product without a separate channel-specific operator.

Use the prediction IDs rather than inventing new expected science after observing output. Interior impulses and sinusoids exercise fixed linear components; donor impulses exercise detector mixing; split-state fixtures exercise recurrence; noncenter causes exercise transitive influence; and odd/even factor enumerations exercise phase-zero identity. These checks are complementary. An interior transfer match cannot close edges or selection, and an output-value match cannot close identity, response, or provenance.

Representation fidelity is a separate record. Reconstruct the exact operator and exceptions from persisted factors, intervals, mappings, and state. Verify that omitted detail is typed optional and inert. A readable plot or plausible header is not proof that the stored representation is complete.

D Informative state and failure checks

Exercise each semantic state independently: not requested, disabled, automatic, rejected, invalid, failed, unavailable, unsupported, and omitted. Also separate original validity, synthesis, replacement, mask membership, numerical finiteness, edge/context, response availability, eligibility, and uncertainty. A fixture that combines every failure into NaN or one generic bit cannot demonstrate these distinctions.

Additionally separate pair-malformed, x-invalid, r-invalid, mapping-unavailable, leakage-unavailable, level-shift-candidate, level-shift-accepted, physical-transition-unmodeled, propagated-transition-influence, additive-offset-applied, additive-offset-unavailable, segment-interior, application-specific plateau-admissibility, plateau-advisory, reset, atmospheric-diagnostic-unavailable, and optional-conditioned-r-unavailable states. Confirm that missing one pair member fails required admission while independent validity of a present member remains expressible.

For each required output member, inject a bounded write or authority failure and verify atomic completion behavior. Inspect that no prior-observation factor, donor, state, mask, rate, product link, or completion marker remains. Then toggle the complete spike-event/donor manifest and other optional inert diagnostics; the scientific and numerical bundle, compact treatment state, and spike-population summary must remain identical apart from the permitted optional detail. Separately verify that selected despiking actually modifies accepted target x cells according to the resolved recovery policy.

Every learned operation requires an additional stop check before apply. The review must bind the declared learning population and every numerical predicate to approved authority, show the complete candidate disposition, and preserve learned-evidence and resolved-plan state. If the resolved plan is absent or incompatible, apply does not begin. A native-cadence result produced after silent fallback is not evidence of an applied learned plan.

Bounded iterative notch refinement requires one such record per attempt. Trace the attempt and current accepted-plan indices, original input, learning parent/population, prior accepted plan, complete proposal, typed disposition, and any new immutable evaluation product, together with candidate dispositions, response and science budgets, and continuation or stop cause. Confirm that the final plan is replayed against the original input unless a separately authorized cascade is recorded. Rejected attempts do not create accepted plans or evaluation products. The record distinguishes actual attempts, configured maximum, and accepted plans; an early stop has no artificial no-op attempts. Maximum-attempt, oscillating, and nonconvergent states are not convergence.

E Informative completion checklist

- Every SCI-RTC-REQ identifier has one explicit disposition and evidence record; ranges are not used as a substitute for per-ID closure.

- Every applicable SCI-RTC-PRED identifier has a preregistered expected result and tolerance, plus observed values and messages.
- Every open owner decision is either resolved by approved authority or blocks the named operation/claim exactly as the ledger states.
- Every RTC application context retains exact paired raw x/r identity and independent validity; conditioned x is the required numerical RTC output, raw r parentage and causal uses remain in the bundle, and SCI-CAL consumes only x while binding the complete consumer-neutral RTC parent.
- Application context, immutable resolved plan, and realized record are distinct one-way objects; context labels neither partition admitted operation classes nor change the atomic bundle schema.
- Missing either pair member fails admission; no reorder, independent join, copy, synthesis, zero fill, r calibration, r-to-x correction, or r-as-x donor route is accepted.
- The owner-modified raw donor ratio is checked in both direction and availability cases, with no dependency on legacy responsivity.
- Direct synthesis/replacement is universally excluded, while noncenter full-support influence is preserved and evaluated by each declared downstream consumer policy.
- Complete response includes every enabled stage or is typed unavailable.
- Leakage diagnostics preserve separate atmosphere and bright-source parents, estimator bias controls, scalar/frequency status, support, uncertainty, and pre/post response; agreement is not automatic promotion.
- Level-shift learning uses original paired data with isolated-spike candidates masked or downweighted; donor replacement begins only after segmentation, remains within stable segments, and never crosses a shift.
- Selected despiking modifies accepted target x cells. Normal output retains compact treatment state and useful spike-population counts and characteristics; a complete event/donor manifest is optional diagnostic detail and its toggle is numerically inert.
- Every accepted shift has paired amplitudes or typed unavailability, transition mask, guards, segment identity, reset/carry state, application-specific plateau support, selected additive correction only with sufficient stable support, persistent transition flags, and no gain- or responsivity-change model.
- Atmospheric templates are diagnostic evidence only and do not alter science x; numerical common-mode removal requires separate authority.
- A conditioned r output is unavailable unless its own operator, response, support, unit, validity, uncertainty, and provenance are bound.
- Requested, effective, observation-resolved, learned-evidence, resolved-plan, realized, and immutable product/stage substates remain distinct within the context–plan–record lifecycle.
- Every covariance or uncertainty claim explicitly states its included and excluded components and correlations; excluded or unknown terms are not silently zero, and only declared complete coverage is called total.
- Each learned operation names its learning population, and apply rejects any mutation of the resolved plan; online adaptation is a separate unavailable or separately authorized estimator.
- Filter validation varies scientific design choices over the admitted scan, projected-beam, and source domain; coefficient fixtures alone do not establish science qualification.
- Iterative notch refinement has a finite configured maximum, distinct actual-attempt and accepted-plan counts, no artificial no-op attempts, complete cumulative proposals, immutable accepted-plan apply, explicit original-input replay or cascade authority, and a typed stop or nonconvergence disposition.
- Required failures are atomic; optional detail is inert; unexpected error-level messages are zero in any run claimed successful.
- The final report names exactly one supported claim layer and explicitly withholds all stronger layers.

This checklist does not itself establish any item. It is a review index for future evidence governed by the shared core.